

# GLOBAL AI COMPANIES & AI ENGINEER CAREER GUIDE

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## 1,000-PAGE COMPREHENSIVE REFERENCE WORKBOOK

Company profiles • Products • Technologies • Skills • Interviews • Career planning

Prepared for learners building practical AI applications, including VoiceRAG AI: a multilingual conversational knowledge assistant with retrieval, evidence checks, citations and voice interaction.

Edition date: September 2026. Hiring pages and requirements change; verify every live vacancy directly with the employer before applying.

**Important: this is an educational reference, not a guarantee of employment or a claim that listed companies are currently hiring for a particular role.**

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## Suggested learning order

Learn Python, SQL, Git and testing first. Build text RAG, then hybrid retrieval and citations. Add voice and multilingual handling after the text system is reliable. Add Neo4j only when graph relationships solve a tested multi-hop question. Deploy, measure, document and publish.

## How to use company profiles

Use each profile to understand product areas and possible engineering problems. Then open the official career page, check the exact location and level, and map requirements to evidence you can demonstrate.

# AI industry map and role families

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## AI Engineer

Integrates model APIs, retrieval, tools, safety checks, monitoring and product workflows. Strong portfolio evidence is a deployed app with tests and measured quality.

## Machine Learning Engineer

Builds data pipelines, model training or fine-tuning, evaluation, serving, monitoring and model lifecycle infrastructure.

## Applied Scientist / Research Engineer

Implements experiments, evaluates hypotheses and may contribute to papers or model development; requirements vary by team.

## Data Engineer

Builds ingestion, transformation, storage, quality checks, data lineage and access controls.

## MLOps / AI Platform Engineer

Builds repeatable deployment, serving, monitoring, evaluation pipelines and infrastructure automation.

## AI Product Engineer

Combines frontend/backend engineering, model APIs, UX, analytics and product iteration.

## AI Solutions Engineer

Maps customer requirements to architecture, prototypes, integrations, deployment constraints and outcomes.

## Research Scientist

Develops scientific methods and hypotheses; many roles expect advanced research experience.

## Shared foundations

Python, SQL, data structures, APIs, Git, testing, debugging, basic statistics, secure configuration, written communication and the ability to explain trade-offs.

## Evidence beats keyword lists

A public repository with tests, an architecture diagram, a working demo, a small evaluation set and an honest limitations section can demonstrate applied skills more clearly than unsupported skill claims.

# Company profile 1: OpenAI

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## Products and business areas

Frontier language and multimodal models, ChatGPT, developer APIs, tool use and enterprise AI.

## Relevant engineering work

Python, APIs, retrieval, evaluation, safety, product engineering and scalable backend services.

## Portfolio project aligned to this company

Build a cited document assistant with tool calling, answer evaluation, cost tracking and abstention tests.

## How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

## Official careers source

<https://openai.com/careers/>

## Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

## Questions to research

What customer or developer problem does OpenAI solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# OpenAI: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of OpenAI's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, APIs, retrieval, evaluation, safety, product engineering and scalable backend services. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a cited document assistant with tool calling, answer evaluation, cost tracking and abstention tests.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 2: Anthropic

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### Products and business areas

Claude models, developer API, enterprise AI and model safety research.

### Relevant engineering work

Python, robust software engineering, evaluations, prompt injection defense and secure system design.

### Portfolio project aligned to this company

Build a RAG assistant with evidence checks, adversarial tests and reproducible evaluations.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.anthropic.com/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Anthropic solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Anthropic: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Anthropic's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, robust software engineering, evaluations, prompt injection defense and secure system design. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a RAG assistant with evidence checks, adversarial tests and reproducible evaluations.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 3: Google DeepMind

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### Products and business areas

Gemini, multimodal AI, scientific machine learning, robotics and foundational research.

### Relevant engineering work

Software engineering, ML fundamentals, data structures, evaluation and distributed systems.

### Portfolio project aligned to this company

Build a retrieval or multimodal prototype and document data, metrics and limitations.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://deepmind.google/careers/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Google DeepMind solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?



# Google DeepMind: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Google DeepMind's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Software engineering, ML fundamentals, data structures, evaluation and distributed systems. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a retrieval or multimodal prototype and document data, metrics and limitations.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 4: Google

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### Products and business areas

Search, cloud AI, Gemini integrations, data infrastructure and developer tools.

### Relevant engineering work

Python/Java/C++, SQL, algorithms, cloud services, monitoring and ML fundamentals.

### Portfolio project aligned to this company

Deploy a retrieval API with tests, observability, access control and cloud architecture.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.google.com/about/careers/applications/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Google solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Google: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Google's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/Java/C++, SQL, algorithms, cloud services, monitoring and ML fundamentals. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Deploy a retrieval API with tests, observability, access control and cloud architecture.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 5: Microsoft

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### Products and business areas

Azure AI, Copilot products, enterprise AI and developer platforms.

### Relevant engineering work

Python/C#, APIs, Azure fundamentals, identity, security, retrieval and enterprise integration.

### Portfolio project aligned to this company

Create a company-policy assistant with permissions, citations, audit logs and evaluations.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://careers.microsoft.com/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Microsoft solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Microsoft: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Microsoft's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/C#, APIs, Azure fundamentals, identity, security, retrieval and enterprise integration. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Create a company-policy assistant with permissions, citations, audit logs and evaluations.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 6: NVIDIA

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### Products and business areas

GPU computing, AI accelerators, inference systems, robotics and AI infrastructure.

### Relevant engineering work

Python/C++, performance profiling, GPU concepts, systems engineering and ML fundamentals.

### Portfolio project aligned to this company

Benchmark an inference workload and report throughput, latency, memory and hardware limits.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.nvidia.com/en-us/about-nvidia/careers/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does NVIDIA solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# NVIDIA: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of NVIDIA's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/C++, performance profiling, GPU concepts, systems engineering and ML fundamentals. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Benchmark an inference workload and report throughput, latency, memory and hardware limits.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 7: Meta

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### Products and business areas

Llama models, recommendations, social products, AI infrastructure and multimodal systems.

### Relevant engineering work

Python/C++, ML, experimentation, data pipelines, systems and responsible deployment.

### Portfolio project aligned to this company

Compare open-model inference settings for quality, latency, resource use and safety.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.metacareers.com/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Meta solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?



# Meta: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Meta's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/C++, ML, experimentation, data pipelines, systems and responsible deployment. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Compare open-model inference settings for quality, latency, resource use and safety.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 8: Amazon / AWS

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### Products and business areas

Cloud infrastructure, AI services, model hosting, recommendation and commerce systems.

### Relevant engineering work

Python, APIs, cloud, IAM, event-driven architecture, monitoring and cost management.

### Portfolio project aligned to this company

Deploy a retrieval service with object storage, a model endpoint, least-privilege access and logs.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.amazon.jobs/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Amazon / AWS solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Amazon / AWS: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Amazon / AWS's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, APIs, cloud, IAM, event-driven architecture, monitoring and cost management. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Deploy a retrieval service with object storage, a model endpoint, least-privilege access and logs.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 9: Apple

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### Products and business areas

On-device intelligence, machine learning, silicon, privacy-aware product experiences.

### Relevant engineering work

Swift/C++/Python depending on role, ML, performance, privacy and platform engineering.

### Portfolio project aligned to this company

Prototype a privacy-conscious local model workflow and describe compute and data boundaries.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://jobs.apple.com/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Apple solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Apple: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Apple's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Swift/C++/Python depending on role, ML, performance, privacy and platform engineering. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Prototype a privacy-conscious local model workflow and describe compute and data boundaries.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 10: IBM

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### Products and business areas

Enterprise AI, hybrid cloud, automation, data platforms and consulting.

### Relevant engineering work

Python, SQL, APIs, enterprise architecture, governance, cloud and stakeholder communication.

### Portfolio project aligned to this company

Create a governed knowledge assistant with source tracking and role-aware retrieval.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.ibm.com/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does IBM solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# IBM: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of IBM's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, SQL, APIs, enterprise architecture, governance, cloud and stakeholder communication. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Create a governed knowledge assistant with source tracking and role-aware retrieval.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

# Company profile 11: Adobe

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## Products and business areas

Creative software, document workflows, digital experience and generative media.

## Relevant engineering work

Python/JavaScript, APIs, data processing, multimodal workflows and evaluation.

## Portfolio project aligned to this company

Build an asset search tool with metadata filters, citations and human review.

## How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

## Official careers source

<https://careers.adobe.com/>

## Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

## Questions to research

What customer or developer problem does Adobe solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?



# Adobe: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Adobe's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/JavaScript, APIs, data processing, multimodal workflows and evaluation. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build an asset search tool with metadata filters, citations and human review.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 12: Salesforce

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### Products and business areas

CRM, enterprise agents, customer workflows, data platforms and business applications.

### Relevant engineering work

APIs, Python/Java, data modeling, permissions, workflow integration and testing.

### Portfolio project aligned to this company

Create a support agent that retrieves policy evidence and asks before taking actions.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://careers.salesforce.com/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Salesforce solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Salesforce: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Salesforce's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

APIs, Python/Java, data modeling, permissions, workflow integration and testing. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Create a support agent that retrieves policy evidence and asks before taking actions.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 13: Databricks

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### Products and business areas

Lakehouse data platform, data engineering, analytics and ML operations.

### Relevant engineering work

Python, SQL, Spark concepts, data quality, pipelines, experiment tracking and deployment.

### Portfolio project aligned to this company

Build an ingestion-to-evaluation pipeline with versioned data and documented metrics.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.databricks.com/company/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Databricks solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Databricks: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Databricks's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, SQL, Spark concepts, data quality, pipelines, experiment tracking and deployment. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build an ingestion-to-evaluation pipeline with versioned data and documented metrics.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 14: Snowflake

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### Products and business areas

Cloud data platform, analytics, data sharing and enterprise AI workloads.

### Relevant engineering work

SQL, Python, data modeling, governance, pipelines and cloud fundamentals.

### Portfolio project aligned to this company

Create an analytics/RAG workflow with data lineage and access policies.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://careers.snowflake.com/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Snowflake solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Snowflake: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Snowflake's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

SQL, Python, data modeling, governance, pipelines and cloud fundamentals. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Create an analytics/RAG workflow with data lineage and access policies.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 15: Cohere

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### Products and business areas

Enterprise language models, retrieval, search and business AI.

### Relevant engineering work

Python, embeddings, retrieval, reranking, API design, privacy and evaluations.

### Portfolio project aligned to this company

Compare dense retrieval, lexical retrieval and reranking on labelled questions.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://cohere.com/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Cohere solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?



# Cohere: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Cohere's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, embeddings, retrieval, reranking, API design, privacy and evaluations. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Compare dense retrieval, lexical retrieval and reranking on labelled questions.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 16: Mistral AI

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### Products and business areas

Language models, open-weight models, inference and enterprise AI services.

### Relevant engineering work

Python, LLM inference, API integration, deployment, evaluation and performance trade-offs.

### Portfolio project aligned to this company

Run a small model locally or through an API and measure quality, speed and cost.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://mistral.ai/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Mistral AI solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Mistral AI: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Mistral AI's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, LLM inference, API integration, deployment, evaluation and performance trade-offs. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Run a small model locally or through an API and measure quality, speed and cost.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 17: Hugging Face

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### Products and business areas

Open model ecosystem, datasets, model hosting, libraries and community tools.

### Relevant engineering work

Python, Transformers, datasets, evaluation, model cards and open-source collaboration.

### Portfolio project aligned to this company

Publish a reproducible model or dataset demo with documentation and limitations.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://huggingface.co/jobs>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Hugging Face solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Hugging Face: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Hugging Face's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, Transformers, datasets, evaluation, model cards and open-source collaboration. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Publish a reproducible model or dataset demo with documentation and limitations.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 18: Perplexity

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### Products and business areas

Answer-oriented search, web retrieval, citations and conversational research.

### Relevant engineering work

Search, retrieval, ranking, Python, web data processing and evaluation.

### Portfolio project aligned to this company

Build a research assistant with URLs, timestamps and claim-to-source mapping.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.perplexity.ai/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Perplexity solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Perplexity: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Perplexity's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Search, retrieval, ranking, Python, web data processing and evaluation. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a research assistant with URLs, timestamps and claim-to-source mapping.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 19: Scale AI

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### Products and business areas

Data infrastructure, model evaluation, data quality and AI deployment services.

### Relevant engineering work

Python, SQL, data quality, annotation design, evaluation and analytics.

### Portfolio project aligned to this company

Create a labelled evaluation dataset with instructions and disagreement handling.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://scale.com/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Scale AI solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?



# Scale AI: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Scale AI's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, SQL, data quality, annotation design, evaluation and analytics. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Create a labelled evaluation dataset with instructions and disagreement handling.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 20: Palantir

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### Products and business areas

Data integration, operational analytics, enterprise platforms and AI workflows.

### Relevant engineering work

Python/Java, data modeling, pipelines, APIs, SQL, deployment and problem solving.

### Portfolio project aligned to this company

Build a traceable data-to-decision dashboard with provenance and permissions.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.palantir.com/careers/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Palantir solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Palantir: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Palantir's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/Java, data modeling, pipelines, APIs, SQL, deployment and problem solving. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a traceable data-to-decision dashboard with provenance and permissions.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 21: Tesla

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### Products and business areas

Autonomy, robotics, manufacturing AI, perception and optimization.

### Relevant engineering work

Python/C++, ML, computer vision, data pipelines, testing and performance.

### Portfolio project aligned to this company

Train a small vision model or simulate an optimization problem and document limitations.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.tesla.com/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Tesla solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Tesla: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Tesla's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/C++, ML, computer vision, data pipelines, testing and performance. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Train a small vision model or simulate an optimization problem and document limitations.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 22: Waymo

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### Products and business areas

Autonomous driving, simulation, perception, prediction and planning.

### Relevant engineering work

Python/C++, ML, geometry, simulation, data engineering and rigorous evaluation.

### Portfolio project aligned to this company

Use a public dataset for a modest perception task with clear metrics and splits.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://waymo.com/careers/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Waymo solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Waymo: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Waymo's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/C++, ML, geometry, simulation, data engineering and rigorous evaluation. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Use a public dataset for a modest perception task with clear metrics and splits.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 23: Uber

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### Products and business areas

Marketplace optimization, mapping, recommendations, forecasting and ML platforms.

### Relevant engineering work

Python, SQL, statistics, experimentation, ML pipelines and monitoring.

### Portfolio project aligned to this company

Build a demand forecast or ranking baseline and explain leakage prevention.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.uber.com/us/en/careers/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Uber solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?



# Uber: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Uber's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, SQL, statistics, experimentation, ML pipelines and monitoring. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a demand forecast or ranking baseline and explain leakage prevention.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 24: Spotify

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### Products and business areas

Recommendations, personalization, audio understanding and experimentation.

### Relevant engineering work

Python, SQL, ranking, embeddings, recommender systems and data pipelines.

### Portfolio project aligned to this company

Build a content-based recommender and compare it with a popularity baseline.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.lifeatspotify.com/jobs>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Spotify solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Spotify: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Spotify's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, SQL, ranking, embeddings, recommender systems and data pipelines. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a content-based recommender and compare it with a popularity baseline.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 25: TikTok / ByteDance

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### Products and business areas

Recommendation systems, content understanding, ads, creative AI and video systems.

### Relevant engineering work

Python/C++, ranking, deep learning, experimentation and streaming data.

### Portfolio project aligned to this company

Build a recommendation prototype with offline metrics and feedback-loop caveats.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://jobs.bytedance.com/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does TikTok / ByteDance solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# TikTok / ByteDance: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of TikTok / ByteDance's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/C++, ranking, deep learning, experimentation and streaming data. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a recommendation prototype with offline metrics and feedback-loop caveats.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 26: Qualcomm

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### Products and business areas

Edge AI, mobile compute, embedded inference and chipsets.

### Relevant engineering work

C/C++, Python, ML deployment, model compression and performance measurement.

### Portfolio project aligned to this company

Quantize a small model and compare size, inference time and accuracy.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.qualcomm.com/company/careers/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Qualcomm solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Qualcomm: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Qualcomm's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

C/C++, Python, ML deployment, model compression and performance measurement. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Quantize a small model and compare size, inference time and accuracy.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 27: Intel

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### Products and business areas

Processors, AI acceleration, edge computing and enterprise systems.

### Relevant engineering work

Python/C++, performance profiling, ML fundamentals, systems and optimization.

### Portfolio project aligned to this company

Compare inference settings on available hardware and document benchmark conditions.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://jobs.intel.com/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Intel solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?



# Intel: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Intel's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/C++, performance profiling, ML fundamentals, systems and optimization. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Compare inference settings on available hardware and document benchmark conditions.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 28: ServiceNow

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### Products and business areas

Enterprise workflow automation, IT service management and AI agents.

### Relevant engineering work

Python/JavaScript, REST APIs, workflow design, access controls and testing.

### Portfolio project aligned to this company

Build a help-desk workflow that retrieves knowledge and requires confirmation before ticket changes.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://careers.servicenow.com/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does ServiceNow solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# ServiceNow: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of ServiceNow's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/JavaScript, REST APIs, workflow design, access controls and testing. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a help-desk workflow that retrieves knowledge and requires confirmation before ticket changes.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 29: SAP

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### Products and business areas

Enterprise resource planning, business data, analytics and enterprise AI.

### Relevant engineering work

SQL, Python/Java, enterprise data models, APIs, governance and process understanding.

### Portfolio project aligned to this company

Build a synthetic procurement assistant that traces answers to source records.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://jobs.sap.com/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does SAP solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# SAP: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of SAP's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

SQL, Python/Java, enterprise data models, APIs, governance and process understanding. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a synthetic procurement assistant that traces answers to source records.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 30: Oracle

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### Products and business areas

Cloud infrastructure, databases, enterprise applications and AI services.

### Relevant engineering work

SQL, Python/Java, databases, APIs, cloud security, performance and monitoring.

### Portfolio project aligned to this company

Create a structured-data question-answering API with query validation and access checks.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.oracle.com/careers/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Oracle solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Oracle: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Oracle's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

SQL, Python/Java, databases, APIs, cloud security, performance and monitoring. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Create a structured-data question-answering API with query validation and access checks.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 31: GitHub

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### Products and business areas

Developer platform, code search, collaboration and AI coding assistance.

### Relevant engineering work

Git, Python/TypeScript, APIs, security, testing and developer experience.

### Portfolio project aligned to this company

Build a repository Q&A tool that cites file paths and line ranges.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://github.com/about/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does GitHub solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?



# GitHub: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of GitHub's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Git, Python/TypeScript, APIs, security, testing and developer experience. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a repository Q&A tool that cites file paths and line ranges.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 32: Cloudflare

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### Products and business areas

Internet infrastructure, security, edge computing and developer services.

### Relevant engineering work

Python/Go/Rust, networking, distributed systems, APIs, security and performance.

### Portfolio project aligned to this company

Deploy a small API with rate limits, safe errors and documented security controls.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.cloudflare.com/careers/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Cloudflare solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Cloudflare: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Cloudflare's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/Go/Rust, networking, distributed systems, APIs, security and performance. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Deploy a small API with rate limits, safe errors and documented security controls.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 33: MongoDB

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### Products and business areas

Document database, developer tools, search and AI data infrastructure.

### Relevant engineering work

Python, data modeling, indexing, APIs, query performance and integration.

### Portfolio project aligned to this company

Store document metadata and sessions with indexes and a documented schema.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.mongodb.com/company/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does MongoDB solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# MongoDB: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of MongoDB's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, data modeling, indexing, APIs, query performance and integration. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Store document metadata and sessions with indexes and a documented schema.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 34: Neo4j

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### Products and business areas

Graph database, connected data, graph analytics and knowledge graph applications.

### Relevant engineering work

Cypher, graph modeling, Python, data ingestion, graph algorithms and APIs.

### Portfolio project aligned to this company

Build an entity graph and demonstrate a useful multi-hop question.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://neo4j.com/careers/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Neo4j solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Neo4j: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Neo4j's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Cypher, graph modeling, Python, data ingestion, graph algorithms and APIs. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build an entity graph and demonstrate a useful multi-hop question.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 35: Pinecone

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### Products and business areas

Managed vector database and retrieval infrastructure.

### Relevant engineering work

Python, embeddings, retrieval evaluation, indexing, metadata filters and APIs.

### Portfolio project aligned to this company

Benchmark retrieval across chunk sizes, filters and vector search settings.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.pinecone.io/careers/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Pinecone solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?



# Pinecone: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Pinecone's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, embeddings, retrieval evaluation, indexing, metadata filters and APIs. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Benchmark retrieval across chunk sizes, filters and vector search settings.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 36: Qdrant

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### Products and business areas

Vector search and retrieval infrastructure.

### Relevant engineering work

Python, embeddings, vector indexing, filtering, deployment and benchmarking.

### Portfolio project aligned to this company

Compare search latency and relevance under different indexing/filter settings.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://qdrant.tech/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Qdrant solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Qdrant: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Qdrant's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, embeddings, vector indexing, filtering, deployment and benchmarking. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Compare search latency and relevance under different indexing/filter settings.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 37: LangChain

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### Products and business areas

Frameworks and tooling for LLM apps, agents and retrieval pipelines.

### Relevant engineering work

Python/TypeScript, LLM APIs, RAG, testing, integrations and software engineering.

### Portfolio project aligned to this company

Build a retrieval chain with component boundaries, traces and automated tests.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.langchain.com/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does LangChain solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# LangChain: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of LangChain's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/TypeScript, LLM APIs, RAG, testing, integrations and software engineering. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a retrieval chain with component boundaries, traces and automated tests.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 38: LlamaIndex

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### Products and business areas

Data connectors, document indexing, retrieval and context augmentation.

### Relevant engineering work

Python, data ingestion, retrieval, indexing strategies, evaluation and APIs.

### Portfolio project aligned to this company

Build an ingestion pipeline with source metadata and incremental updates.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.llamaindex.ai/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does LlamaIndex solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# LlamaIndex: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of LlamaIndex's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, data ingestion, retrieval, indexing strategies, evaluation and APIs. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build an ingestion pipeline with source metadata and incremental updates.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 39: Weights & Biases

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### Products and business areas

Experiment tracking, model evaluation, observability and ML workflows.

### Relevant engineering work

Python, experiment design, metrics, reproducibility, visualization and lifecycle practices.

### Portfolio project aligned to this company

Track RAG experiments with versioned datasets, settings and comparable metrics.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://wandb.ai/site/careers/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Weights & Biases solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?



# Weights & Biases: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Weights & Biases's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, experiment design, metrics, reproducibility, visualization and lifecycle practices. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Track RAG experiments with versioned datasets, settings and comparable metrics.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 40: ElevenLabs

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### Products and business areas

Speech synthesis, voice tools, audio generation and speech interfaces.

### Relevant engineering work

Python, audio APIs, streaming, latency, UX and speech evaluation.

### Portfolio project aligned to this company

Add spoken answers to RAG while retaining a visible transcript and interruption handling.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://elevenlabs.io/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does ElevenLabs solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# ElevenLabs: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of ElevenLabs's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, audio APIs, streaming, latency, UX and speech evaluation. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Add spoken answers to RAG while retaining a visible transcript and interruption handling.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 41: Deepgram

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### Products and business areas

Speech recognition, audio intelligence and voice application infrastructure.

### Relevant engineering work

Python, streaming APIs, audio processing, latency and transcript evaluation.

### Portfolio project aligned to this company

Measure transcription errors on a small test set and allow transcript correction.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://deepgram.com/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Deepgram solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Deepgram: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Deepgram's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, streaming APIs, audio processing, latency and transcript evaluation. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Measure transcription errors on a small test set and allow transcript correction.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 42: Vercel

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### Products and business areas

Frontend deployment, developer experience and AI application tooling.

### Relevant engineering work

TypeScript, React, APIs, frontend performance, deployment and observability.

### Portfolio project aligned to this company

Deploy a chat UI with streaming responses, accessible controls and loading states.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://vercel.com/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Vercel solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Vercel: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Vercel's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

TypeScript, React, APIs, frontend performance, deployment and observability. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Deploy a chat UI with streaming responses, accessible controls and loading states.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 43: Replit

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### Products and business areas

Cloud development environments, coding agents and software creation tools.

### Relevant engineering work

TypeScript/Python, developer tools, agents, execution environments and reliability.

### Portfolio project aligned to this company

Build a constrained code helper that explains changes and requires approval before execution.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://replit.com/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Replit solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?



# Replit: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Replit's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

TypeScript/Python, developer tools, agents, execution environments and reliability. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a constrained code helper that explains changes and requires approval before execution.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 44: Duolingo

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### Products and business areas

Language learning, personalization, speech and educational AI.

### Relevant engineering work

Python, experimentation, ranking, speech/text processing, analytics and product metrics.

### Portfolio project aligned to this company

Build a language practice assistant that tracks learning goals and uncertainty.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://careers.duolingo.com/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Duolingo solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Duolingo: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Duolingo's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, experimentation, ranking, speech/text processing, analytics and product metrics. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a language practice assistant that tracks learning goals and uncertainty.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 45: Stripe

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### Products and business areas

Payments infrastructure, fraud systems, developer APIs and financial operations.

### Relevant engineering work

Python/Ruby/Java, API design, reliability, data systems, security and technical writing.

### Portfolio project aligned to this company

Build a simulated support assistant that protects sensitive data and cites policy rules.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://stripe.com/jobs>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Stripe solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Stripe: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Stripe's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/Ruby/Java, API design, reliability, data systems, security and technical writing. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a simulated support assistant that protects sensitive data and cites policy rules.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 46: JPMorgan Chase

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### Products and business areas

Financial services, risk analytics, AI engineering and enterprise automation.

### Relevant engineering work

Python, SQL, software engineering, data governance, model risk and secure deployment.

### Portfolio project aligned to this company

Build a public-disclosure assistant with citations and strict data provenance.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://careers.jpmorgan.com/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does JPMorgan Chase solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# JPMorgan Chase: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of JPMorgan Chase's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, SQL, software engineering, data governance, model risk and secure deployment. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a public-disclosure assistant with citations and strict data provenance.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 47: Bloomberg

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### Products and business areas

Financial data, search, analytics, NLP and real-time information systems.

### Relevant engineering work

Python/C++, SQL, data quality, NLP, search relevance and reliable systems.

### Portfolio project aligned to this company

Build entity-linked news search using public data and measure retrieval accuracy.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.bloomberg.com/company/careers/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Bloomberg solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?



# Bloomberg: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Bloomberg's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/C++, SQL, data quality, NLP, search relevance and reliable systems. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build entity-linked news search using public data and measure retrieval accuracy.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 48: Siemens

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### Products and business areas

Industrial automation, digital twins, manufacturing and applied AI.

### Relevant engineering work

Python/C++, time-series data, industrial systems, testing, edge deployment and domain work.

### Portfolio project aligned to this company

Build a synthetic sensor anomaly detector and analyze false positives.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://jobs.siemens.com/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Siemens solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Siemens: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Siemens's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/C++, time-series data, industrial systems, testing, edge deployment and domain work. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Build a synthetic sensor anomaly detector and analyze false positives.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 49: Bosch

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### Products and business areas

Automotive, industrial IoT, embedded systems, computer vision and applied AI.

### Relevant engineering work

Python/C++, ML, sensor data, embedded constraints, validation and safety engineering.

### Portfolio project aligned to this company

Prototype predictive maintenance using public or synthetic sensor data.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.bosch.com/careers/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Bosch solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Bosch: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Bosch's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/C++, ML, sensor data, embedded constraints, validation and safety engineering. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Prototype predictive maintenance using public or synthetic sensor data.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 50: Nokia

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### Products and business areas

Telecommunications, networks, optimization and applied machine learning.

### Relevant engineering work

Python/C++, time-series analytics, networking, distributed systems and testing.

### Portfolio project aligned to this company

Analyze synthetic network KPIs and produce a transparent anomaly report.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.nokia.com/about-us/careers/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Nokia solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Nokia: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Nokia's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/C++, time-series analytics, networking, distributed systems and testing. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Analyze synthetic network KPIs and produce a transparent anomaly report.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 51: Red Hat

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### Products and business areas

Enterprise Linux, Kubernetes, automation and open-source AI infrastructure.

### Relevant engineering work

Linux, Python, containers, Kubernetes concepts, CI/CD, security and collaboration.

### Portfolio project aligned to this company

Containerize an AI service with health checks, secret handling and recovery notes.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.redhat.com/en/jobs/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Red Hat solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?



# Red Hat: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Red Hat's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Linux, Python, containers, Kubernetes concepts, CI/CD, security and collaboration. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Containerize an AI service with health checks, secret handling and recovery notes.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 52: Elastic

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### Products and business areas

Search, observability, security analytics and vector retrieval.

### Relevant engineering work

Python, search relevance, indexing, APIs, distributed systems and monitoring.

### Portfolio project aligned to this company

Compare lexical and vector search on exact-match and semantic queries.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.elastic.co/about/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Elastic solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Elastic: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Elastic's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python, search relevance, indexing, APIs, distributed systems and monitoring. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Compare lexical and vector search on exact-match and semantic queries.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 53: Groq

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### Products and business areas

AI inference hardware and low-latency model serving.

### Relevant engineering work

Python/C++, systems, benchmarking, API integration and performance analysis.

### Portfolio project aligned to this company

Measure time-to-first-token and total response time across prompt lengths.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://groq.com/careers/>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does Groq solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# Groq: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of Groq's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Python/C++, systems, benchmarking, API integration and performance analysis. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Measure time-to-first-token and total response time across prompt lengths.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

## Company profile 54: CoreWeave

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### Products and business areas

GPU cloud infrastructure and AI compute.

### Relevant engineering work

Linux, Python, containers, orchestration, cloud networking and observability.

### Portfolio project aligned to this company

Deploy a GPU-aware service where available, with a documented CPU fallback.

### How to investigate a role

Use the official career page. Filter by internship, new graduate, entry-level, software, ML, data, AI platform or applied AI as appropriate. Compare must-have requirements with your real experience; do not infer an opening from this guide.

### Official careers source

<https://www.coreweave.com/careers>

### Application evidence checklist

Can I explain the team problem? Can I show related code or a measured experiment? Can I explain trade-offs, failure modes and security? Can I tailor my resume without exaggerating?

### Questions to research

What customer or developer problem does CoreWeave solve? Which product area uses AI? Which skills are repeated in current job descriptions? What would a realistic first project look like?

# CoreWeave: interview and portfolio worksheet

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## Explain the company in your own words

Write a 60-second summary of CoreWeave's relevant products, likely users, and one engineering challenge. Validate the description against official product documentation.

## Skills mapping

Linux, Python, containers, orchestration, cloud networking and observability. Map each requirement to a course, code sample, test, project, or honest learning plan.

## Project evidence

Deploy a GPU-aware service where available, with a documented CPU fallback.

## Questions for the recruiter or team

What is the scope of this role? Which product team would I join? What does success look like in the first 90 days? Which skills are required on day one versus learned on the job?

## Application checklist

Verify location, eligibility, level, deadline and role status. Customize the resume summary and project bullets to the role. Check all claims and links before submitting.

# Skill roadmap workbook 1: Python and software engineering

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## Learning focus

Functions, classes, exceptions, modules, type hints, packaging, virtual environments, unit tests, logging, linting, HTTP, JSON, retries and timeouts.

## Practical task

Choose one small component of VoiceRAG AI related to python and software engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?



## Skill roadmap workbook 2: SQL and data fundamentals

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### Learning focus

Joins, grouping, window functions, CTEs, indexing, schema design, data quality, duplicates, missing values, freshness and lineage.

### Practical task

Choose one small component of VoiceRAG AI related to sql and data fundamentals. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 3: Machine learning foundations

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### Learning focus

Train/validation/test splits, leakage, baselines, classification, regression, precision, recall, F1, ROC-AUC, calibration, bias and distribution shift.

### Practical task

Choose one small component of VoiceRAG AI related to machine learning foundations. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 4: LLM application engineering

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### Learning focus

Tokens, context windows, structured outputs, tool calling, streaming, rate limits, retries, prompt injection, safe failure behavior and cost control.

### Practical task

Choose one small component of VoiceRAG AI related to llm application engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 5: RAG and retrieval

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## Learning focus

Parsing, chunking, metadata, embeddings, BM25, vector search, hybrid fusion, reranking, citation mapping, abstention and retrieval evaluation.

## Practical task

Choose one small component of VoiceRAG AI related to rag and retrieval. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 6: Voice and multilingual systems

---

### Learning focus

Raw transcript, language detection, query normalization, code-switching, speech errors, streaming, turn-taking, interruption and latency.

### Practical task

Choose one small component of VoiceRAG AI related to voice and multilingual systems. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 7: Deployment and operations

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## Learning focus

Docker, configuration, secret management, authentication, rate limiting, TLS, health checks, logs, traces, CI/CD, rollback and cost monitoring.

## Practical task

Choose one small component of VoiceRAG AI related to deployment and operations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 8: Python and software engineering

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### Learning focus

Functions, classes, exceptions, modules, type hints, packaging, virtual environments, unit tests, logging, linting, HTTP, JSON, retries and timeouts.

### Practical task

Choose one small component of VoiceRAG AI related to python and software engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 9: SQL and data fundamentals

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### Learning focus

Joins, grouping, window functions, CTEs, indexing, schema design, data quality, duplicates, missing values, freshness and lineage.

### Practical task

Choose one small component of VoiceRAG AI related to sql and data fundamentals. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?



# Skill roadmap workbook 10: Machine learning foundations

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## Learning focus

Train/validation/test splits, leakage, baselines, classification, regression, precision, recall, F1, ROC-AUC, calibration, bias and distribution shift.

## Practical task

Choose one small component of VoiceRAG AI related to machine learning foundations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 11: LLM application engineering

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## Learning focus

Tokens, context windows, structured outputs, tool calling, streaming, rate limits, retries, prompt injection, safe failure behavior and cost control.

## Practical task

Choose one small component of VoiceRAG AI related to llm application engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 12: RAG and retrieval

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### Learning focus

Parsing, chunking, metadata, embeddings, BM25, vector search, hybrid fusion, reranking, citation mapping, abstention and retrieval evaluation.

### Practical task

Choose one small component of VoiceRAG AI related to rag and retrieval. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 13: Voice and multilingual systems

---

### Learning focus

Raw transcript, language detection, query normalization, code-switching, speech errors, streaming, turn-taking, interruption and latency.

### Practical task

Choose one small component of VoiceRAG AI related to voice and multilingual systems. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 14: Deployment and operations

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## Learning focus

Docker, configuration, secret management, authentication, rate limiting, TLS, health checks, logs, traces, CI/CD, rollback and cost monitoring.

## Practical task

Choose one small component of VoiceRAG AI related to deployment and operations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 15: Python and software engineering

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## Learning focus

Functions, classes, exceptions, modules, type hints, packaging, virtual environments, unit tests, logging, linting, HTTP, JSON, retries and timeouts.

## Practical task

Choose one small component of VoiceRAG AI related to python and software engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 16: SQL and data fundamentals

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### Learning focus

Joins, grouping, window functions, CTEs, indexing, schema design, data quality, duplicates, missing values, freshness and lineage.

### Practical task

Choose one small component of VoiceRAG AI related to sql and data fundamentals. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 17: Machine learning foundations

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## Learning focus

Train/validation/test splits, leakage, baselines, classification, regression, precision, recall, F1, ROC-AUC, calibration, bias and distribution shift.

## Practical task

Choose one small component of VoiceRAG AI related to machine learning foundations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?



# Skill roadmap workbook 18: LLM application engineering

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## Learning focus

Tokens, context windows, structured outputs, tool calling, streaming, rate limits, retries, prompt injection, safe failure behavior and cost control.

## Practical task

Choose one small component of VoiceRAG AI related to llm application engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 19: RAG and retrieval

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### Learning focus

Parsing, chunking, metadata, embeddings, BM25, vector search, hybrid fusion, reranking, citation mapping, abstention and retrieval evaluation.

### Practical task

Choose one small component of VoiceRAG AI related to rag and retrieval. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 20: Voice and multilingual systems

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### Learning focus

Raw transcript, language detection, query normalization, code-switching, speech errors, streaming, turn-taking, interruption and latency.

### Practical task

Choose one small component of VoiceRAG AI related to voice and multilingual systems. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 21: Deployment and operations

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### Learning focus

Docker, configuration, secret management, authentication, rate limiting, TLS, health checks, logs, traces, CI/CD, rollback and cost monitoring.

### Practical task

Choose one small component of VoiceRAG AI related to deployment and operations. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 22: Python and software engineering

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### Learning focus

Functions, classes, exceptions, modules, type hints, packaging, virtual environments, unit tests, logging, linting, HTTP, JSON, retries and timeouts.

### Practical task

Choose one small component of VoiceRAG AI related to python and software engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 23: SQL and data fundamentals

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### Learning focus

Joins, grouping, window functions, CTEs, indexing, schema design, data quality, duplicates, missing values, freshness and lineage.

### Practical task

Choose one small component of VoiceRAG AI related to sql and data fundamentals. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 24: Machine learning foundations

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### Learning focus

Train/validation/test splits, leakage, baselines, classification, regression, precision, recall, F1, ROC-AUC, calibration, bias and distribution shift.

### Practical task

Choose one small component of VoiceRAG AI related to machine learning foundations. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 25: LLM application engineering

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### Learning focus

Tokens, context windows, structured outputs, tool calling, streaming, rate limits, retries, prompt injection, safe failure behavior and cost control.

### Practical task

Choose one small component of VoiceRAG AI related to llm application engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?



## Skill roadmap workbook 26: RAG and retrieval

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### Learning focus

Parsing, chunking, metadata, embeddings, BM25, vector search, hybrid fusion, reranking, citation mapping, abstention and retrieval evaluation.

### Practical task

Choose one small component of VoiceRAG AI related to rag and retrieval. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 27: Voice and multilingual systems

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### Learning focus

Raw transcript, language detection, query normalization, code-switching, speech errors, streaming, turn-taking, interruption and latency.

### Practical task

Choose one small component of VoiceRAG AI related to voice and multilingual systems. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 28: Deployment and operations

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### Learning focus

Docker, configuration, secret management, authentication, rate limiting, TLS, health checks, logs, traces, CI/CD, rollback and cost monitoring.

### Practical task

Choose one small component of VoiceRAG AI related to deployment and operations. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 29: Python and software engineering

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### Learning focus

Functions, classes, exceptions, modules, type hints, packaging, virtual environments, unit tests, logging, linting, HTTP, JSON, retries and timeouts.

### Practical task

Choose one small component of VoiceRAG AI related to python and software engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 30: SQL and data fundamentals

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### Learning focus

Joins, grouping, window functions, CTEs, indexing, schema design, data quality, duplicates, missing values, freshness and lineage.

### Practical task

Choose one small component of VoiceRAG AI related to sql and data fundamentals. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 31: Machine learning foundations

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## Learning focus

Train/validation/test splits, leakage, baselines, classification, regression, precision, recall, F1, ROC-AUC, calibration, bias and distribution shift.

## Practical task

Choose one small component of VoiceRAG AI related to machine learning foundations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 32: LLM application engineering

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### Learning focus

Tokens, context windows, structured outputs, tool calling, streaming, rate limits, retries, prompt injection, safe failure behavior and cost control.

### Practical task

Choose one small component of VoiceRAG AI related to llm application engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 33: RAG and retrieval

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### Learning focus

Parsing, chunking, metadata, embeddings, BM25, vector search, hybrid fusion, reranking, citation mapping, abstention and retrieval evaluation.

### Practical task

Choose one small component of VoiceRAG AI related to rag and retrieval. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?



## Skill roadmap workbook 34: Voice and multilingual systems

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### Learning focus

Raw transcript, language detection, query normalization, code-switching, speech errors, streaming, turn-taking, interruption and latency.

### Practical task

Choose one small component of VoiceRAG AI related to voice and multilingual systems. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 35: Deployment and operations

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## Learning focus

Docker, configuration, secret management, authentication, rate limiting, TLS, health checks, logs, traces, CI/CD, rollback and cost monitoring.

## Practical task

Choose one small component of VoiceRAG AI related to deployment and operations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 36: Python and software engineering

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## Learning focus

Functions, classes, exceptions, modules, type hints, packaging, virtual environments, unit tests, logging, linting, HTTP, JSON, retries and timeouts.

## Practical task

Choose one small component of VoiceRAG AI related to python and software engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 37: SQL and data fundamentals

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### Learning focus

Joins, grouping, window functions, CTEs, indexing, schema design, data quality, duplicates, missing values, freshness and lineage.

### Practical task

Choose one small component of VoiceRAG AI related to sql and data fundamentals. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 38: Machine learning foundations

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### Learning focus

Train/validation/test splits, leakage, baselines, classification, regression, precision, recall, F1, ROC-AUC, calibration, bias and distribution shift.

### Practical task

Choose one small component of VoiceRAG AI related to machine learning foundations. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 39: LLM application engineering

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### Learning focus

Tokens, context windows, structured outputs, tool calling, streaming, rate limits, retries, prompt injection, safe failure behavior and cost control.

### Practical task

Choose one small component of VoiceRAG AI related to llm application engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 40: RAG and retrieval

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### Learning focus

Parsing, chunking, metadata, embeddings, BM25, vector search, hybrid fusion, reranking, citation mapping, abstention and retrieval evaluation.

### Practical task

Choose one small component of VoiceRAG AI related to rag and retrieval. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 41: Voice and multilingual systems

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## Learning focus

Raw transcript, language detection, query normalization, code-switching, speech errors, streaming, turn-taking, interruption and latency.

## Practical task

Choose one small component of VoiceRAG AI related to voice and multilingual systems. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?



## Skill roadmap workbook 42: Deployment and operations

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### Learning focus

Docker, configuration, secret management, authentication, rate limiting, TLS, health checks, logs, traces, CI/CD, rollback and cost monitoring.

### Practical task

Choose one small component of VoiceRAG AI related to deployment and operations. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 43: Python and software engineering

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### Learning focus

Functions, classes, exceptions, modules, type hints, packaging, virtual environments, unit tests, logging, linting, HTTP, JSON, retries and timeouts.

### Practical task

Choose one small component of VoiceRAG AI related to python and software engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 44: SQL and data fundamentals

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### Learning focus

Joins, grouping, window functions, CTEs, indexing, schema design, data quality, duplicates, missing values, freshness and lineage.

### Practical task

Choose one small component of VoiceRAG AI related to sql and data fundamentals. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 45: Machine learning foundations

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## Learning focus

Train/validation/test splits, leakage, baselines, classification, regression, precision, recall, F1, ROC-AUC, calibration, bias and distribution shift.

## Practical task

Choose one small component of VoiceRAG AI related to machine learning foundations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 46: LLM application engineering

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### Learning focus

Tokens, context windows, structured outputs, tool calling, streaming, rate limits, retries, prompt injection, safe failure behavior and cost control.

### Practical task

Choose one small component of VoiceRAG AI related to llm application engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 47: RAG and retrieval

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## Learning focus

Parsing, chunking, metadata, embeddings, BM25, vector search, hybrid fusion, reranking, citation mapping, abstention and retrieval evaluation.

## Practical task

Choose one small component of VoiceRAG AI related to rag and retrieval. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 48: Voice and multilingual systems

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## Learning focus

Raw transcript, language detection, query normalization, code-switching, speech errors, streaming, turn-taking, interruption and latency.

## Practical task

Choose one small component of VoiceRAG AI related to voice and multilingual systems. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 49: Deployment and operations

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## Learning focus

Docker, configuration, secret management, authentication, rate limiting, TLS, health checks, logs, traces, CI/CD, rollback and cost monitoring.

## Practical task

Choose one small component of VoiceRAG AI related to deployment and operations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?



# Skill roadmap workbook 50: Python and software engineering

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## Learning focus

Functions, classes, exceptions, modules, type hints, packaging, virtual environments, unit tests, logging, linting, HTTP, JSON, retries and timeouts.

## Practical task

Choose one small component of VoiceRAG AI related to python and software engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 51: SQL and data fundamentals

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## Learning focus

Joins, grouping, window functions, CTEs, indexing, schema design, data quality, duplicates, missing values, freshness and lineage.

## Practical task

Choose one small component of VoiceRAG AI related to sql and data fundamentals. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 52: Machine learning foundations

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### Learning focus

Train/validation/test splits, leakage, baselines, classification, regression, precision, recall, F1, ROC-AUC, calibration, bias and distribution shift.

### Practical task

Choose one small component of VoiceRAG AI related to machine learning foundations. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 53: LLM application engineering

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### Learning focus

Tokens, context windows, structured outputs, tool calling, streaming, rate limits, retries, prompt injection, safe failure behavior and cost control.

### Practical task

Choose one small component of VoiceRAG AI related to llm application engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 54: RAG and retrieval

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## Learning focus

Parsing, chunking, metadata, embeddings, BM25, vector search, hybrid fusion, reranking, citation mapping, abstention and retrieval evaluation.

## Practical task

Choose one small component of VoiceRAG AI related to rag and retrieval. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 55: Voice and multilingual systems

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## Learning focus

Raw transcript, language detection, query normalization, code-switching, speech errors, streaming, turn-taking, interruption and latency.

## Practical task

Choose one small component of VoiceRAG AI related to voice and multilingual systems. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 56: Deployment and operations

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## Learning focus

Docker, configuration, secret management, authentication, rate limiting, TLS, health checks, logs, traces, CI/CD, rollback and cost monitoring.

## Practical task

Choose one small component of VoiceRAG AI related to deployment and operations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 57: Python and software engineering

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## Learning focus

Functions, classes, exceptions, modules, type hints, packaging, virtual environments, unit tests, logging, linting, HTTP, JSON, retries and timeouts.

## Practical task

Choose one small component of VoiceRAG AI related to python and software engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?



## Skill roadmap workbook 58: SQL and data fundamentals

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### Learning focus

Joins, grouping, window functions, CTEs, indexing, schema design, data quality, duplicates, missing values, freshness and lineage.

### Practical task

Choose one small component of VoiceRAG AI related to sql and data fundamentals. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 59: Machine learning foundations

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### Learning focus

Train/validation/test splits, leakage, baselines, classification, regression, precision, recall, F1, ROC-AUC, calibration, bias and distribution shift.

### Practical task

Choose one small component of VoiceRAG AI related to machine learning foundations. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 60: LLM application engineering

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## Learning focus

Tokens, context windows, structured outputs, tool calling, streaming, rate limits, retries, prompt injection, safe failure behavior and cost control.

## Practical task

Choose one small component of VoiceRAG AI related to llm application engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 61: RAG and retrieval

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## Learning focus

Parsing, chunking, metadata, embeddings, BM25, vector search, hybrid fusion, reranking, citation mapping, abstention and retrieval evaluation.

## Practical task

Choose one small component of VoiceRAG AI related to rag and retrieval. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 62: Voice and multilingual systems

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### Learning focus

Raw transcript, language detection, query normalization, code-switching, speech errors, streaming, turn-taking, interruption and latency.

### Practical task

Choose one small component of VoiceRAG AI related to voice and multilingual systems. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 63: Deployment and operations

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## Learning focus

Docker, configuration, secret management, authentication, rate limiting, TLS, health checks, logs, traces, CI/CD, rollback and cost monitoring.

## Practical task

Choose one small component of VoiceRAG AI related to deployment and operations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 64: Python and software engineering

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## Learning focus

Functions, classes, exceptions, modules, type hints, packaging, virtual environments, unit tests, logging, linting, HTTP, JSON, retries and timeouts.

## Practical task

Choose one small component of VoiceRAG AI related to python and software engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 65: SQL and data fundamentals

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## Learning focus

Joins, grouping, window functions, CTEs, indexing, schema design, data quality, duplicates, missing values, freshness and lineage.

## Practical task

Choose one small component of VoiceRAG AI related to sql and data fundamentals. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?



# Skill roadmap workbook 66: Machine learning foundations

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## Learning focus

Train/validation/test splits, leakage, baselines, classification, regression, precision, recall, F1, ROC-AUC, calibration, bias and distribution shift.

## Practical task

Choose one small component of VoiceRAG AI related to machine learning foundations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 67: LLM application engineering

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## Learning focus

Tokens, context windows, structured outputs, tool calling, streaming, rate limits, retries, prompt injection, safe failure behavior and cost control.

## Practical task

Choose one small component of VoiceRAG AI related to llm application engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 68: RAG and retrieval

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### Learning focus

Parsing, chunking, metadata, embeddings, BM25, vector search, hybrid fusion, reranking, citation mapping, abstention and retrieval evaluation.

### Practical task

Choose one small component of VoiceRAG AI related to rag and retrieval. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 69: Voice and multilingual systems

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## Learning focus

Raw transcript, language detection, query normalization, code-switching, speech errors, streaming, turn-taking, interruption and latency.

## Practical task

Choose one small component of VoiceRAG AI related to voice and multilingual systems. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 70: Deployment and operations

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## Learning focus

Docker, configuration, secret management, authentication, rate limiting, TLS, health checks, logs, traces, CI/CD, rollback and cost monitoring.

## Practical task

Choose one small component of VoiceRAG AI related to deployment and operations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 71: Python and software engineering

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## Learning focus

Functions, classes, exceptions, modules, type hints, packaging, virtual environments, unit tests, logging, linting, HTTP, JSON, retries and timeouts.

## Practical task

Choose one small component of VoiceRAG AI related to python and software engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 72: SQL and data fundamentals

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### Learning focus

Joins, grouping, window functions, CTEs, indexing, schema design, data quality, duplicates, missing values, freshness and lineage.

### Practical task

Choose one small component of VoiceRAG AI related to sql and data fundamentals. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 73: Machine learning foundations

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### Learning focus

Train/validation/test splits, leakage, baselines, classification, regression, precision, recall, F1, ROC-AUC, calibration, bias and distribution shift.

### Practical task

Choose one small component of VoiceRAG AI related to machine learning foundations. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?



# Skill roadmap workbook 74: LLM application engineering

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## Learning focus

Tokens, context windows, structured outputs, tool calling, streaming, rate limits, retries, prompt injection, safe failure behavior and cost control.

## Practical task

Choose one small component of VoiceRAG AI related to llm application engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 75: RAG and retrieval

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## Learning focus

Parsing, chunking, metadata, embeddings, BM25, vector search, hybrid fusion, reranking, citation mapping, abstention and retrieval evaluation.

## Practical task

Choose one small component of VoiceRAG AI related to rag and retrieval. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 76: Voice and multilingual systems

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## Learning focus

Raw transcript, language detection, query normalization, code-switching, speech errors, streaming, turn-taking, interruption and latency.

## Practical task

Choose one small component of VoiceRAG AI related to voice and multilingual systems. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 77: Deployment and operations

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## Learning focus

Docker, configuration, secret management, authentication, rate limiting, TLS, health checks, logs, traces, CI/CD, rollback and cost monitoring.

## Practical task

Choose one small component of VoiceRAG AI related to deployment and operations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 78: Python and software engineering

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## Learning focus

Functions, classes, exceptions, modules, type hints, packaging, virtual environments, unit tests, logging, linting, HTTP, JSON, retries and timeouts.

## Practical task

Choose one small component of VoiceRAG AI related to python and software engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 79: SQL and data fundamentals

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### Learning focus

Joins, grouping, window functions, CTEs, indexing, schema design, data quality, duplicates, missing values, freshness and lineage.

### Practical task

Choose one small component of VoiceRAG AI related to sql and data fundamentals. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 80: Machine learning foundations

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## Learning focus

Train/validation/test splits, leakage, baselines, classification, regression, precision, recall, F1, ROC-AUC, calibration, bias and distribution shift.

## Practical task

Choose one small component of VoiceRAG AI related to machine learning foundations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 81: LLM application engineering

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## Learning focus

Tokens, context windows, structured outputs, tool calling, streaming, rate limits, retries, prompt injection, safe failure behavior and cost control.

## Practical task

Choose one small component of VoiceRAG AI related to llm application engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?



## Skill roadmap workbook 82: RAG and retrieval

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### Learning focus

Parsing, chunking, metadata, embeddings, BM25, vector search, hybrid fusion, reranking, citation mapping, abstention and retrieval evaluation.

### Practical task

Choose one small component of VoiceRAG AI related to rag and retrieval. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 83: Voice and multilingual systems

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### Learning focus

Raw transcript, language detection, query normalization, code-switching, speech errors, streaming, turn-taking, interruption and latency.

### Practical task

Choose one small component of VoiceRAG AI related to voice and multilingual systems. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 84: Deployment and operations

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## Learning focus

Docker, configuration, secret management, authentication, rate limiting, TLS, health checks, logs, traces, CI/CD, rollback and cost monitoring.

## Practical task

Choose one small component of VoiceRAG AI related to deployment and operations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 85: Python and software engineering

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## Learning focus

Functions, classes, exceptions, modules, type hints, packaging, virtual environments, unit tests, logging, linting, HTTP, JSON, retries and timeouts.

## Practical task

Choose one small component of VoiceRAG AI related to python and software engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 86: SQL and data fundamentals

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### Learning focus

Joins, grouping, window functions, CTEs, indexing, schema design, data quality, duplicates, missing values, freshness and lineage.

### Practical task

Choose one small component of VoiceRAG AI related to sql and data fundamentals. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 87: Machine learning foundations

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## Learning focus

Train/validation/test splits, leakage, baselines, classification, regression, precision, recall, F1, ROC-AUC, calibration, bias and distribution shift.

## Practical task

Choose one small component of VoiceRAG AI related to machine learning foundations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 88: LLM application engineering

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### Learning focus

Tokens, context windows, structured outputs, tool calling, streaming, rate limits, retries, prompt injection, safe failure behavior and cost control.

### Practical task

Choose one small component of VoiceRAG AI related to llm application engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 89: RAG and retrieval

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## Learning focus

Parsing, chunking, metadata, embeddings, BM25, vector search, hybrid fusion, reranking, citation mapping, abstention and retrieval evaluation.

## Practical task

Choose one small component of VoiceRAG AI related to rag and retrieval. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?



# Skill roadmap workbook 90: Voice and multilingual systems

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## Learning focus

Raw transcript, language detection, query normalization, code-switching, speech errors, streaming, turn-taking, interruption and latency.

## Practical task

Choose one small component of VoiceRAG AI related to voice and multilingual systems. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 91: Deployment and operations

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## Learning focus

Docker, configuration, secret management, authentication, rate limiting, TLS, health checks, logs, traces, CI/CD, rollback and cost monitoring.

## Practical task

Choose one small component of VoiceRAG AI related to deployment and operations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 92: Python and software engineering

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### Learning focus

Functions, classes, exceptions, modules, type hints, packaging, virtual environments, unit tests, logging, linting, HTTP, JSON, retries and timeouts.

### Practical task

Choose one small component of VoiceRAG AI related to python and software engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

## Skill roadmap workbook 93: SQL and data fundamentals

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### Learning focus

Joins, grouping, window functions, CTEs, indexing, schema design, data quality, duplicates, missing values, freshness and lineage.

### Practical task

Choose one small component of VoiceRAG AI related to sql and data fundamentals. Define its inputs, outputs, expected behavior and failure modes before coding.

### Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

### Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

### Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 94: Machine learning foundations

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## Learning focus

Train/validation/test splits, leakage, baselines, classification, regression, precision, recall, F1, ROC-AUC, calibration, bias and distribution shift.

## Practical task

Choose one small component of VoiceRAG AI related to machine learning foundations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 95: LLM application engineering

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## Learning focus

Tokens, context windows, structured outputs, tool calling, streaming, rate limits, retries, prompt injection, safe failure behavior and cost control.

## Practical task

Choose one small component of VoiceRAG AI related to llm application engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 96: RAG and retrieval

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## Learning focus

Parsing, chunking, metadata, embeddings, BM25, vector search, hybrid fusion, reranking, citation mapping, abstention and retrieval evaluation.

## Practical task

Choose one small component of VoiceRAG AI related to rag and retrieval. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 97: Voice and multilingual systems

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## Learning focus

Raw transcript, language detection, query normalization, code-switching, speech errors, streaming, turn-taking, interruption and latency.

## Practical task

Choose one small component of VoiceRAG AI related to voice and multilingual systems. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?



# Skill roadmap workbook 98: Deployment and operations

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## Learning focus

Docker, configuration, secret management, authentication, rate limiting, TLS, health checks, logs, traces, CI/CD, rollback and cost monitoring.

## Practical task

Choose one small component of VoiceRAG AI related to deployment and operations. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 99: Python and software engineering

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## Learning focus

Functions, classes, exceptions, modules, type hints, packaging, virtual environments, unit tests, logging, linting, HTTP, JSON, retries and timeouts.

## Practical task

Choose one small component of VoiceRAG AI related to python and software engineering. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# Skill roadmap workbook 100: SQL and data fundamentals

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## Learning focus

Joins, grouping, window functions, CTEs, indexing, schema design, data quality, duplicates, missing values, freshness and lineage.

## Practical task

Choose one small component of VoiceRAG AI related to sql and data fundamentals. Define its inputs, outputs, expected behavior and failure modes before coding.

## Acceptance criteria

Include a reproducible run command, one positive test, one edge-case test, one failure test and a short note explaining what the result does and does not prove.

## Interview explanation

Explain the concept in plain language, give a practical example, describe one trade-off and mention how you would test it.

## Review prompts

What can go wrong? What data is sensitive? Which metric matters? What fallback should the user see? What did you actually measure?

# VoiceRAG AI architecture workbook 1: Milestone 1 — Local text RAG

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## Design focus

Ingest public or self-created PDFs; extract text; preserve page, heading and document metadata; split into chunks; retrieve candidates; display source snippets.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 2: Milestone 2 — Grounded ans

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### Design focus

Pass retrieved evidence to the model, require source IDs, validate citation IDs and return insufficient-evidence responses when the corpus does not support an answer.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 3: Milestone 3 — Hybrid retrieval

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### Design focus

Combine BM25 and vector search. Use a documented fusion method. Add a reranker only after a baseline exists and measure relevance gains against added latency.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 4: Milestone 4 — Evaluation

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## Design focus

Create a labelled set of direct, multi-hop, exact-ID, unanswerable and multilingual questions. Measure retrieval recall, citation support, correctness, abstention, latency and cost.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 5: Milestone 5 — Voice

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## Design focus

Add speech recognition and text-to-speech. Display raw transcript separately from normalized query and answer. Allow users to correct transcript errors.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.



# VoiceRAG AI architecture workbook 6: Milestone 6 — Multilingual s

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## Design focus

Start with a declared language set. Test accents, names, technical vocabulary, numerals and code-switching. Do not claim support for untested languages.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 7: Milestone 7 — Knowledge gr

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## Design focus

Add Neo4j only if explicit entity relationships and multi-hop traversal help. Retain source provenance and compare against the no-graph baseline.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 8: Milestone 8 — Deployment

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## Design focus

Containerize the app, store secrets outside Git, configure health checks, limit upload size, add timeouts and document public-demo limitations.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 9: Security and privacy

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## Design focus

Treat retrieved content as untrusted. Restrict file types and sizes, isolate users' documents, protect credentials, avoid logging secrets and define deletion behavior.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 10: Monitoring and cost

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## Design focus

Track latency percentiles, errors, retrieval failures, provider outages, token use, cost and resource utilization. Separate fixed hosting costs from per-request costs.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 11: Milestone 1 — Local text RA

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## Design focus

Ingest public or self-created PDFs; extract text; preserve page, heading and document metadata; split into chunks; retrieve candidates; display source snippets.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 12: Milestone 2 — Grounded an

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## Design focus

Pass retrieved evidence to the model, require source IDs, validate citation IDs and return insufficient-evidence responses when the corpus does not support an answer.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 13: Milestone 3 — Hybrid retrieval

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## Design focus

Combine BM25 and vector search. Use a documented fusion method. Add a reranker only after a baseline exists and measure relevance gains against added latency.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.



# VoiceRAG AI architecture workbook 14: Milestone 4 — Evaluation

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## Design focus

Create a labelled set of direct, multi-hop, exact-ID, unanswerable and multilingual questions. Measure retrieval recall, citation support, correctness, abstention, latency and cost.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 15: Milestone 5 — Voice

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### Design focus

Add speech recognition and text-to-speech. Display raw transcript separately from normalized query and answer. Allow users to correct transcript errors.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 16: Milestone 6 — Multilingual s

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## Design focus

Start with a declared language set. Test accents, names, technical vocabulary, numerals and code-switching. Do not claim support for untested languages.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 17: Milestone 7 — Knowledge g

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## Design focus

Add Neo4j only if explicit entity relationships and multi-hop traversal help. Retain source provenance and compare against the no-graph baseline.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 18: Milestone 8 — Deployment

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## Design focus

Containerize the app, store secrets outside Git, configure health checks, limit upload size, add timeouts and document public-demo limitations.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 19: Security and privacy

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## Design focus

Treat retrieved content as untrusted. Restrict file types and sizes, isolate users' documents, protect credentials, avoid logging secrets and define deletion behavior.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 20: Monitoring and cost

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### Design focus

Track latency percentiles, errors, retrieval failures, provider outages, token use, cost and resource utilization. Separate fixed hosting costs from per-request costs.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 21: Milestone 1 — Local text RA

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## Design focus

Ingest public or self-created PDFs; extract text; preserve page, heading and document metadata; split into chunks; retrieve candidates; display source snippets.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.



## VoiceRAG AI architecture workbook 22: Milestone 2 — Grounded an

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### Design focus

Pass retrieved evidence to the model, require source IDs, validate citation IDs and return insufficient-evidence responses when the corpus does not support an answer.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 23: Milestone 3 — Hybrid retrieval

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### Design focus

Combine BM25 and vector search. Use a documented fusion method. Add a reranker only after a baseline exists and measure relevance gains against added latency.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 24: Milestone 4 — Evaluation

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## Design focus

Create a labelled set of direct, multi-hop, exact-ID, unanswerable and multilingual questions. Measure retrieval recall, citation support, correctness, abstention, latency and cost.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 25: Milestone 5 — Voice

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### Design focus

Add speech recognition and text-to-speech. Display raw transcript separately from normalized query and answer. Allow users to correct transcript errors.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 26: Milestone 6 — Multilingual s

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## Design focus

Start with a declared language set. Test accents, names, technical vocabulary, numerals and code-switching. Do not claim support for untested languages.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 27: Milestone 7 — Knowledge g

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## Design focus

Add Neo4j only if explicit entity relationships and multi-hop traversal help. Retain source provenance and compare against the no-graph baseline.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 28: Milestone 8 — Deployment

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## Design focus

Containerize the app, store secrets outside Git, configure health checks, limit upload size, add timeouts and document public-demo limitations.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 29: Security and privacy

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## Design focus

Treat retrieved content as untrusted. Restrict file types and sizes, isolate users' documents, protect credentials, avoid logging secrets and define deletion behavior.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.



# VoiceRAG AI architecture workbook 30: Monitoring and cost

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## Design focus

Track latency percentiles, errors, retrieval failures, provider outages, token use, cost and resource utilization. Separate fixed hosting costs from per-request costs.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 31: Milestone 1 — Local text RA

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## Design focus

Ingest public or self-created PDFs; extract text; preserve page, heading and document metadata; split into chunks; retrieve candidates; display source snippets.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 32: Milestone 2 — Grounded an

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### Design focus

Pass retrieved evidence to the model, require source IDs, validate citation IDs and return insufficient-evidence responses when the corpus does not support an answer.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 33: Milestone 3 — Hybrid retrieval

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## Design focus

Combine BM25 and vector search. Use a documented fusion method. Add a reranker only after a baseline exists and measure relevance gains against added latency.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 34: Milestone 4 — Evaluation

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## Design focus

Create a labelled set of direct, multi-hop, exact-ID, unanswerable and multilingual questions. Measure retrieval recall, citation support, correctness, abstention, latency and cost.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 35: Milestone 5 — Voice

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### Design focus

Add speech recognition and text-to-speech. Display raw transcript separately from normalized query and answer. Allow users to correct transcript errors.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 36: Milestone 6 — Multilingual s

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## Design focus

Start with a declared language set. Test accents, names, technical vocabulary, numerals and code-switching. Do not claim support for untested languages.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 37: Milestone 7 — Knowledge g

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## Design focus

Add Neo4j only if explicit entity relationships and multi-hop traversal help. Retain source provenance and compare against the no-graph baseline.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.



# VoiceRAG AI architecture workbook 38: Milestone 8 — Deployment

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## Design focus

Containerize the app, store secrets outside Git, configure health checks, limit upload size, add timeouts and document public-demo limitations.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 39: Security and privacy

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### Design focus

Treat retrieved content as untrusted. Restrict file types and sizes, isolate users' documents, protect credentials, avoid logging secrets and define deletion behavior.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 40: Monitoring and cost

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### Design focus

Track latency percentiles, errors, retrieval failures, provider outages, token use, cost and resource utilization. Separate fixed hosting costs from per-request costs.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 41: Milestone 1 — Local text RA

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## Design focus

Ingest public or self-created PDFs; extract text; preserve page, heading and document metadata; split into chunks; retrieve candidates; display source snippets.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 42: Milestone 2 — Grounded an

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### Design focus

Pass retrieved evidence to the model, require source IDs, validate citation IDs and return insufficient-evidence responses when the corpus does not support an answer.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 43: Milestone 3 — Hybrid retrieval

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## Design focus

Combine BM25 and vector search. Use a documented fusion method. Add a reranker only after a baseline exists and measure relevance gains against added latency.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 44: Milestone 4 — Evaluation

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## Design focus

Create a labelled set of direct, multi-hop, exact-ID, unanswerable and multilingual questions. Measure retrieval recall, citation support, correctness, abstention, latency and cost.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 45: Milestone 5 — Voice

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## Design focus

Add speech recognition and text-to-speech. Display raw transcript separately from normalized query and answer. Allow users to correct transcript errors.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.



# VoiceRAG AI architecture workbook 46: Milestone 6 — Multilingual s

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## Design focus

Start with a declared language set. Test accents, names, technical vocabulary, numerals and code-switching. Do not claim support for untested languages.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 47: Milestone 7 — Knowledge g

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## Design focus

Add Neo4j only if explicit entity relationships and multi-hop traversal help. Retain source provenance and compare against the no-graph baseline.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 48: Milestone 8 — Deployment

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## Design focus

Containerize the app, store secrets outside Git, configure health checks, limit upload size, add timeouts and document public-demo limitations.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 49: Security and privacy

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## Design focus

Treat retrieved content as untrusted. Restrict file types and sizes, isolate users' documents, protect credentials, avoid logging secrets and define deletion behavior.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 50: Monitoring and cost

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## Design focus

Track latency percentiles, errors, retrieval failures, provider outages, token use, cost and resource utilization. Separate fixed hosting costs from per-request costs.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 51: Milestone 1 — Local text RA

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## Design focus

Ingest public or self-created PDFs; extract text; preserve page, heading and document metadata; split into chunks; retrieve candidates; display source snippets.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 52: Milestone 2 — Grounded an

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### Design focus

Pass retrieved evidence to the model, require source IDs, validate citation IDs and return insufficient-evidence responses when the corpus does not support an answer.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 53: Milestone 3 — Hybrid retrieval

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### Design focus

Combine BM25 and vector search. Use a documented fusion method. Add a reranker only after a baseline exists and measure relevance gains against added latency.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.



# VoiceRAG AI architecture workbook 54: Milestone 4 — Evaluation

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## Design focus

Create a labelled set of direct, multi-hop, exact-ID, unanswerable and multilingual questions. Measure retrieval recall, citation support, correctness, abstention, latency and cost.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 55: Milestone 5 — Voice

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### Design focus

Add speech recognition and text-to-speech. Display raw transcript separately from normalized query and answer. Allow users to correct transcript errors.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 56: Milestone 6 — Multilingual s

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## Design focus

Start with a declared language set. Test accents, names, technical vocabulary, numerals and code-switching. Do not claim support for untested languages.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 57: Milestone 7 — Knowledge g

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## Design focus

Add Neo4j only if explicit entity relationships and multi-hop traversal help. Retain source provenance and compare against the no-graph baseline.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 58: Milestone 8 — Deployment

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## Design focus

Containerize the app, store secrets outside Git, configure health checks, limit upload size, add timeouts and document public-demo limitations.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 59: Security and privacy

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## Design focus

Treat retrieved content as untrusted. Restrict file types and sizes, isolate users' documents, protect credentials, avoid logging secrets and define deletion behavior.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 60: Monitoring and cost

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## Design focus

Track latency percentiles, errors, retrieval failures, provider outages, token use, cost and resource utilization. Separate fixed hosting costs from per-request costs.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 61: Milestone 1 — Local text RA

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## Design focus

Ingest public or self-created PDFs; extract text; preserve page, heading and document metadata; split into chunks; retrieve candidates; display source snippets.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.



## VoiceRAG AI architecture workbook 62: Milestone 2 — Grounded an

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### Design focus

Pass retrieved evidence to the model, require source IDs, validate citation IDs and return insufficient-evidence responses when the corpus does not support an answer.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 63: Milestone 3 — Hybrid retrieval

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## Design focus

Combine BM25 and vector search. Use a documented fusion method. Add a reranker only after a baseline exists and measure relevance gains against added latency.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 64: Milestone 4 — Evaluation

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## Design focus

Create a labelled set of direct, multi-hop, exact-ID, unanswerable and multilingual questions. Measure retrieval recall, citation support, correctness, abstention, latency and cost.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 65: Milestone 5 — Voice

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## Design focus

Add speech recognition and text-to-speech. Display raw transcript separately from normalized query and answer. Allow users to correct transcript errors.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 66: Milestone 6 — Multilingual s

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## Design focus

Start with a declared language set. Test accents, names, technical vocabulary, numerals and code-switching. Do not claim support for untested languages.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 67: Milestone 7 — Knowledge g

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## Design focus

Add Neo4j only if explicit entity relationships and multi-hop traversal help. Retain source provenance and compare against the no-graph baseline.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 68: Milestone 8 — Deployment

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## Design focus

Containerize the app, store secrets outside Git, configure health checks, limit upload size, add timeouts and document public-demo limitations.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 69: Security and privacy

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## Design focus

Treat retrieved content as untrusted. Restrict file types and sizes, isolate users' documents, protect credentials, avoid logging secrets and define deletion behavior.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.



# VoiceRAG AI architecture workbook 70: Monitoring and cost

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## Design focus

Track latency percentiles, errors, retrieval failures, provider outages, token use, cost and resource utilization. Separate fixed hosting costs from per-request costs.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 71: Milestone 1 — Local text RA

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## Design focus

Ingest public or self-created PDFs; extract text; preserve page, heading and document metadata; split into chunks; retrieve candidates; display source snippets.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 72: Milestone 2 — Grounded an

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### Design focus

Pass retrieved evidence to the model, require source IDs, validate citation IDs and return insufficient-evidence responses when the corpus does not support an answer.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 73: Milestone 3 — Hybrid retrieval

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## Design focus

Combine BM25 and vector search. Use a documented fusion method. Add a reranker only after a baseline exists and measure relevance gains against added latency.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 74: Milestone 4 — Evaluation

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## Design focus

Create a labelled set of direct, multi-hop, exact-ID, unanswerable and multilingual questions. Measure retrieval recall, citation support, correctness, abstention, latency and cost.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

## VoiceRAG AI architecture workbook 75: Milestone 5 — Voice

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### Design focus

Add speech recognition and text-to-speech. Display raw transcript separately from normalized query and answer. Allow users to correct transcript errors.

### Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

### Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

### Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 76: Milestone 6 — Multilingual s

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## Design focus

Start with a declared language set. Test accents, names, technical vocabulary, numerals and code-switching. Do not claim support for untested languages.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 77: Milestone 7 — Knowledge g

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## Design focus

Add Neo4j only if explicit entity relationships and multi-hop traversal help. Retain source provenance and compare against the no-graph baseline.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.



# VoiceRAG AI architecture workbook 78: Milestone 8 — Deployment

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## Design focus

Containerize the app, store secrets outside Git, configure health checks, limit upload size, add timeouts and document public-demo limitations.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 79: Security and privacy

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## Design focus

Treat retrieved content as untrusted. Restrict file types and sizes, isolate users' documents, protect credentials, avoid logging secrets and define deletion behavior.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# VoiceRAG AI architecture workbook 80: Monitoring and cost

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## Design focus

Track latency percentiles, errors, retrieval failures, provider outages, token use, cost and resource utilization. Separate fixed hosting costs from per-request costs.

## Design question

What is the smallest implementation that proves this component works? What would make it fail? What evidence would demonstrate improvement?

## Implementation notes

Keep interfaces small. Version datasets and configuration. Record source metadata. Never place API keys in client code or a public repository.

## Evaluation plan

Use a fixed question set and baseline. Report both successful cases and failures. Avoid claims such as 100% accurate, hallucination-free or real-time unless the relevant property is measured and qualified.

# Interview preparation 1: Python

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## Practice question

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Expand your answer

Apply the answer to python. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

## What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

## Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 2: SQL

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### Practice question

Why combine BM25 and vector search?

### Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

### Expand your answer

Apply the answer to sql. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 3: Machine learning

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### Practice question

How do you reduce unsupported answers?

### Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

### Expand your answer

Apply the answer to machine learning. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 4: LLMs

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### Practice question

What is reranking?

### Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

### Expand your answer

Apply the answer to llms. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 5: RAG

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### Practice question

When would you use Neo4j?

### Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

### Expand your answer

Apply the answer to rag. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.



## Interview preparation 6: Agents

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### Practice question

How do you protect an API key?

### Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

### Expand your answer

Apply the answer to agents. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 7: Deployment

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### Practice question

How would you evaluate a RAG system?

### Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

### Expand your answer

Apply the answer to deployment. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 8: System design

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### Practice question

How do you handle model provider failure?

### Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

### Expand your answer

Apply the answer to system design. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 9: Voice systems

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### Practice question

How would you test an AI agent?

### Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

### Expand your answer

Apply the answer to voice systems. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 10: Behavioral

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### Practice question

How do you explain a project to a nontechnical person?

### Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

### Expand your answer

Apply the answer to behavioral. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

# Interview preparation 11: Python

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## Practice question

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Expand your answer

Apply the answer to python. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

## What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

## Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 12: SQL

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### Practice question

Why combine BM25 and vector search?

### Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

### Expand your answer

Apply the answer to sql. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 13: Machine learning

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### Practice question

How do you reduce unsupported answers?

### Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

### Expand your answer

Apply the answer to machine learning. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.



## Interview preparation 14: LLMs

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### Practice question

What is reranking?

### Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

### Expand your answer

Apply the answer to llms. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 15: RAG

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### Practice question

When would you use Neo4j?

### Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

### Expand your answer

Apply the answer to rag. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 16: Agents

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### Practice question

How do you protect an API key?

### Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

### Expand your answer

Apply the answer to agents. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 17: Deployment

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### Practice question

How would you evaluate a RAG system?

### Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

### Expand your answer

Apply the answer to deployment. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 18: System design

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### Practice question

How do you handle model provider failure?

### Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

### Expand your answer

Apply the answer to system design. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 19: Voice systems

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### Practice question

How would you test an AI agent?

### Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

### Expand your answer

Apply the answer to voice systems. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 20: Behavioral

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### Practice question

How do you explain a project to a nontechnical person?

### Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

### Expand your answer

Apply the answer to behavioral. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 21: Python

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### Practice question

What is RAG?

### Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

### Expand your answer

Apply the answer to python. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.



## Interview preparation 22: SQL

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### Practice question

Why combine BM25 and vector search?

### Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

### Expand your answer

Apply the answer to sql. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 23: Machine learning

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### Practice question

How do you reduce unsupported answers?

### Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

### Expand your answer

Apply the answer to machine learning. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 24: LLMs

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### Practice question

What is reranking?

### Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

### Expand your answer

Apply the answer to llms. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 25: RAG

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### Practice question

When would you use Neo4j?

### Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

### Expand your answer

Apply the answer to rag. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 26: Agents

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### Practice question

How do you protect an API key?

### Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

### Expand your answer

Apply the answer to agents. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 27: Deployment

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### Practice question

How would you evaluate a RAG system?

### Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

### Expand your answer

Apply the answer to deployment. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 28: System design

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### Practice question

How do you handle model provider failure?

### Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

### Expand your answer

Apply the answer to system design. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 29: Voice systems

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### Practice question

How would you test an AI agent?

### Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

### Expand your answer

Apply the answer to voice systems. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.



## Interview preparation 30: Behavioral

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### Practice question

How do you explain a project to a nontechnical person?

### Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

### Expand your answer

Apply the answer to behavioral. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

# Interview preparation 31: Python

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## Practice question

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Expand your answer

Apply the answer to python. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

## What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

## Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 32: SQL

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### Practice question

Why combine BM25 and vector search?

### Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

### Expand your answer

Apply the answer to sql. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 33: Machine learning

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### Practice question

How do you reduce unsupported answers?

### Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

### Expand your answer

Apply the answer to machine learning. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 34: LLMs

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### Practice question

What is reranking?

### Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

### Expand your answer

Apply the answer to llms. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 35: RAG

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### Practice question

When would you use Neo4j?

### Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

### Expand your answer

Apply the answer to rag. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 36: Agents

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### Practice question

How do you protect an API key?

### Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

### Expand your answer

Apply the answer to agents. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 37: Deployment

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### Practice question

How would you evaluate a RAG system?

### Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

### Expand your answer

Apply the answer to deployment. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.



## Interview preparation 38: System design

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### Practice question

How do you handle model provider failure?

### Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

### Expand your answer

Apply the answer to system design. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 39: Voice systems

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### Practice question

How would you test an AI agent?

### Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

### Expand your answer

Apply the answer to voice systems. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 40: Behavioral

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### Practice question

How do you explain a project to a nontechnical person?

### Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

### Expand your answer

Apply the answer to behavioral. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

# Interview preparation 41: Python

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## Practice question

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Expand your answer

Apply the answer to python. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

## What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

## Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 42: SQL

---

### Practice question

Why combine BM25 and vector search?

### Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

### Expand your answer

Apply the answer to sql. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 43: Machine learning

---

### Practice question

How do you reduce unsupported answers?

### Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

### Expand your answer

Apply the answer to machine learning. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 44: LLMs

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### Practice question

What is reranking?

### Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

### Expand your answer

Apply the answer to llms. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 45: RAG

---

### Practice question

When would you use Neo4j?

### Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

### Expand your answer

Apply the answer to rag. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.



## Interview preparation 46: Agents

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### Practice question

How do you protect an API key?

### Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

### Expand your answer

Apply the answer to agents. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 47: Deployment

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### Practice question

How would you evaluate a RAG system?

### Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

### Expand your answer

Apply the answer to deployment. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 48: System design

---

### Practice question

How do you handle model provider failure?

### Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

### Expand your answer

Apply the answer to system design. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 49: Voice systems

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### Practice question

How would you test an AI agent?

### Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

### Expand your answer

Apply the answer to voice systems. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 50: Behavioral

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### Practice question

How do you explain a project to a nontechnical person?

### Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

### Expand your answer

Apply the answer to behavioral. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

# Interview preparation 51: Python

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## Practice question

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Expand your answer

Apply the answer to python. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

## What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

## Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 52: SQL

---

### Practice question

Why combine BM25 and vector search?

### Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

### Expand your answer

Apply the answer to sql. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 53: Machine learning

---

### Practice question

How do you reduce unsupported answers?

### Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

### Expand your answer

Apply the answer to machine learning. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.



## Interview preparation 54: LLMs

---

### Practice question

What is reranking?

### Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

### Expand your answer

Apply the answer to llms. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 55: RAG

---

### Practice question

When would you use Neo4j?

### Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

### Expand your answer

Apply the answer to rag. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 56: Agents

---

### Practice question

How do you protect an API key?

### Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

### Expand your answer

Apply the answer to agents. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 57: Deployment

---

### Practice question

How would you evaluate a RAG system?

### Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

### Expand your answer

Apply the answer to deployment. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 58: System design

---

### Practice question

How do you handle model provider failure?

### Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

### Expand your answer

Apply the answer to system design. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 59: Voice systems

---

### Practice question

How would you test an AI agent?

### Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

### Expand your answer

Apply the answer to voice systems. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 60: Behavioral

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### Practice question

How do you explain a project to a nontechnical person?

### Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

### Expand your answer

Apply the answer to behavioral. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

# Interview preparation 61: Python

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## Practice question

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Expand your answer

Apply the answer to python. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

## What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

## Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.



## Interview preparation 62: SQL

---

### Practice question

Why combine BM25 and vector search?

### Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

### Expand your answer

Apply the answer to sql. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 63: Machine learning

---

### Practice question

How do you reduce unsupported answers?

### Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

### Expand your answer

Apply the answer to machine learning. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 64: LLMs

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### Practice question

What is reranking?

### Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

### Expand your answer

Apply the answer to llms. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 65: RAG

---

### Practice question

When would you use Neo4j?

### Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

### Expand your answer

Apply the answer to rag. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 66: Agents

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### Practice question

How do you protect an API key?

### Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

### Expand your answer

Apply the answer to agents. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 67: Deployment

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### Practice question

How would you evaluate a RAG system?

### Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

### Expand your answer

Apply the answer to deployment. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 68: System design

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### Practice question

How do you handle model provider failure?

### Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

### Expand your answer

Apply the answer to system design. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 69: Voice systems

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### Practice question

How would you test an AI agent?

### Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

### Expand your answer

Apply the answer to voice systems. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.



## Interview preparation 70: Behavioral

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### Practice question

How do you explain a project to a nontechnical person?

### Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

### Expand your answer

Apply the answer to behavioral. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

# Interview preparation 71: Python

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## Practice question

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Expand your answer

Apply the answer to python. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

## What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

## Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 72: SQL

---

### Practice question

Why combine BM25 and vector search?

### Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

### Expand your answer

Apply the answer to sql. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 73: Machine learning

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### Practice question

How do you reduce unsupported answers?

### Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

### Expand your answer

Apply the answer to machine learning. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 74: LLMs

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### Practice question

What is reranking?

### Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

### Expand your answer

Apply the answer to llms. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 75: RAG

---

### Practice question

When would you use Neo4j?

### Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

### Expand your answer

Apply the answer to rag. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 76: Agents

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### Practice question

How do you protect an API key?

### Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

### Expand your answer

Apply the answer to agents. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 77: Deployment

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### Practice question

How would you evaluate a RAG system?

### Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

### Expand your answer

Apply the answer to deployment. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.



## Interview preparation 78: System design

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### Practice question

How do you handle model provider failure?

### Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

### Expand your answer

Apply the answer to system design. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 79: Voice systems

---

### Practice question

How would you test an AI agent?

### Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

### Expand your answer

Apply the answer to voice systems. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 80: Behavioral

---

### Practice question

How do you explain a project to a nontechnical person?

### Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

### Expand your answer

Apply the answer to behavioral. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

# Interview preparation 81: Python

---

## Practice question

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Expand your answer

Apply the answer to python. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

## What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

## Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 82: SQL

---

### Practice question

Why combine BM25 and vector search?

### Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

### Expand your answer

Apply the answer to sql. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 83: Machine learning

---

### Practice question

How do you reduce unsupported answers?

### Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

### Expand your answer

Apply the answer to machine learning. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 84: LLMs

---

### Practice question

What is reranking?

### Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

### Expand your answer

Apply the answer to llms. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 85: RAG

---

### Practice question

When would you use Neo4j?

### Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

### Expand your answer

Apply the answer to rag. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.



## Interview preparation 86: Agents

---

### Practice question

How do you protect an API key?

### Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

### Expand your answer

Apply the answer to agents. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 87: Deployment

---

### Practice question

How would you evaluate a RAG system?

### Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

### Expand your answer

Apply the answer to deployment. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 88: System design

---

### Practice question

How do you handle model provider failure?

### Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

### Expand your answer

Apply the answer to system design. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 89: Voice systems

---

### Practice question

How would you test an AI agent?

### Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

### Expand your answer

Apply the answer to voice systems. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 90: Behavioral

---

### Practice question

How do you explain a project to a nontechnical person?

### Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

### Expand your answer

Apply the answer to behavioral. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

# Interview preparation 91: Python

---

## Practice question

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Expand your answer

Apply the answer to python. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

## What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

## Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 92: SQL

---

### Practice question

Why combine BM25 and vector search?

### Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

### Expand your answer

Apply the answer to sql. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 93: Machine learning

---

### Practice question

How do you reduce unsupported answers?

### Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

### Expand your answer

Apply the answer to machine learning. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.



## Interview preparation 94: LLMs

---

### Practice question

What is reranking?

### Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

### Expand your answer

Apply the answer to llms. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 95: RAG

---

### Practice question

When would you use Neo4j?

### Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

### Expand your answer

Apply the answer to rag. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 96: Agents

---

### Practice question

How do you protect an API key?

### Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

### Expand your answer

Apply the answer to agents. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 97: Deployment

---

### Practice question

How would you evaluate a RAG system?

### Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

### Expand your answer

Apply the answer to deployment. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 98: System design

---

### Practice question

How do you handle model provider failure?

### Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

### Expand your answer

Apply the answer to system design. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

## Interview preparation 99: Voice systems

---

### Practice question

How would you test an AI agent?

### Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

### Expand your answer

Apply the answer to voice systems. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

### What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

### Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

# Interview preparation 100: Behavioral

---

## Practice question

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Expand your answer

Apply the answer to behavioral. State assumptions, give an example, discuss alternatives, describe evaluation and name at least one failure mode.

## What interviewers may probe

Why did you choose this design? What happens when input is missing? How would you test correctness? What would you monitor after deployment? What would you change at larger scale?

## Self-review

Answer the exact question first. Separate implemented features from planned features. Use measured evidence, not invented metrics. Keep the explanation structured and concise.

# Practical lab 1: PDF parsing

---

## Objective

Extract text and page metadata from public PDFs. Test empty, malformed and scanned documents; record limitations.

## Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

## Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

## Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

## Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?



## Practical lab 2: Chunk metadata

---

### Objective

Create stable chunk IDs with document, page, heading and index. Ensure every result traces to its source.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 3: Baseline retrieval

---

### Objective

Implement lexical retrieval on a fixed question set and report recall@k and missed cases.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 4: Vector retrieval

---

### Objective

Compare semantic search with lexical search using the same questions and relevance labels.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 5: Hybrid fusion

---

### Objective

Combine lexical and dense rankings; compare exact-term and semantic query types.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 6: Reranking

---

### Objective

Rerank candidate passages and measure relevance gains alongside latency and cost.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 7: Answer grounding

---

### Objective

Generate answers only from retrieved passages and validate citation IDs against known sources.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 8: Abstention

---

### Objective

Test questions that the corpus cannot answer; avoid fabricated facts or citations.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 9: Voice transcript

---

### Objective

Display raw speech transcript separately from normalized query and generated answer.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?



## Practical lab 10: Multilingual tests

---

### Objective

Test supported languages, code-switching, names, numbers and technical vocabulary.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 11: Knowledge graph

---

### Objective

Define entity types, relationship types and source provenance; validate extracted edges.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 12: Neo4j query

---

### Objective

Implement one multi-hop Cypher query and compare results against the no-graph baseline.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 13: API reliability

---

### Objective

Add validation, timeouts, structured errors, health checks and tests.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 14: Containerization

---

### Objective

Build a Docker image with external configuration and reproducible run instructions.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 15: Security review

---

### Objective

Review uploads, prompt injection, secrets, access control, logs and deletion policy.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 16: Evaluation report

---

### Objective

Maintain labelled cases and report retrieval quality, citation support, latency and cost.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 17: Cost estimate

---

### Objective

Separate fixed hosting costs from per-request model costs and document assumptions.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?



## Practical lab 18: Model card

---

### Objective

Describe intended use, data sources, evaluation, limitations and failure cases.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 19: Demo video

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### Objective

Show a real query, retrieved evidence, cited answer, an unanswerable query and recovery.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 20: Resume evidence

---

### Objective

Map project features to real code, tests, metrics and your individual contribution.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 21: PDF parsing

---

### Objective

Extract text and page metadata from public PDFs. Test empty, malformed and scanned documents; record limitations.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 22: Chunk metadata

---

### Objective

Create stable chunk IDs with document, page, heading and index. Ensure every result traces to its source.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 23: Baseline retrieval

---

### Objective

Implement lexical retrieval on a fixed question set and report recall@k and missed cases.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 24: Vector retrieval

---

### Objective

Compare semantic search with lexical search using the same questions and relevance labels.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 25: Hybrid fusion

---

### Objective

Combine lexical and dense rankings; compare exact-term and semantic query types.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?



## Practical lab 26: Reranking

---

### Objective

Rerank candidate passages and measure relevance gains alongside latency and cost.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 27: Answer grounding

---

### Objective

Generate answers only from retrieved passages and validate citation IDs against known sources.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 28: Abstention

---

### Objective

Test questions that the corpus cannot answer; avoid fabricated facts or citations.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 29: Voice transcript

---

### Objective

Display raw speech transcript separately from normalized query and generated answer.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 30: Multilingual tests

---

### Objective

Test supported languages, code-switching, names, numbers and technical vocabulary.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 31: Knowledge graph

---

### Objective

Define entity types, relationship types and source provenance; validate extracted edges.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 32: Neo4j query

---

### Objective

Implement one multi-hop Cypher query and compare results against the no-graph baseline.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 33: API reliability

---

### Objective

Add validation, timeouts, structured errors, health checks and tests.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?



## Practical lab 34: Containerization

---

### Objective

Build a Docker image with external configuration and reproducible run instructions.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 35: Security review

---

### Objective

Review uploads, prompt injection, secrets, access control, logs and deletion policy.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 36: Evaluation report

---

### Objective

Maintain labelled cases and report retrieval quality, citation support, latency and cost.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 37: Cost estimate

---

### Objective

Separate fixed hosting costs from per-request model costs and document assumptions.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 38: Model card

---

### Objective

Describe intended use, data sources, evaluation, limitations and failure cases.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 39: Demo video

---

### Objective

Show a real query, retrieved evidence, cited answer, an unanswerable query and recovery.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 40: Resume evidence

---

### Objective

Map project features to real code, tests, metrics and your individual contribution.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

# Practical lab 41: PDF parsing

---

## Objective

Extract text and page metadata from public PDFs. Test empty, malformed and scanned documents; record limitations.

## Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

## Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

## Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

## Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?



## Practical lab 42: Chunk metadata

---

### Objective

Create stable chunk IDs with document, page, heading and index. Ensure every result traces to its source.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 43: Baseline retrieval

---

### Objective

Implement lexical retrieval on a fixed question set and report recall@k and missed cases.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 44: Vector retrieval

---

### Objective

Compare semantic search with lexical search using the same questions and relevance labels.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 45: Hybrid fusion

---

### Objective

Combine lexical and dense rankings; compare exact-term and semantic query types.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 46: Reranking

---

### Objective

Rerank candidate passages and measure relevance gains alongside latency and cost.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 47: Answer grounding

---

### Objective

Generate answers only from retrieved passages and validate citation IDs against known sources.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 48: Abstention

---

### Objective

Test questions that the corpus cannot answer; avoid fabricated facts or citations.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 49: Voice transcript

---

### Objective

Display raw speech transcript separately from normalized query and generated answer.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?



## Practical lab 50: Multilingual tests

---

### Objective

Test supported languages, code-switching, names, numbers and technical vocabulary.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 51: Knowledge graph

---

### Objective

Define entity types, relationship types and source provenance; validate extracted edges.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 52: Neo4j query

---

### Objective

Implement one multi-hop Cypher query and compare results against the no-graph baseline.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 53: API reliability

---

### Objective

Add validation, timeouts, structured errors, health checks and tests.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 54: Containerization

---

### Objective

Build a Docker image with external configuration and reproducible run instructions.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 55: Security review

---

### Objective

Review uploads, prompt injection, secrets, access control, logs and deletion policy.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 56: Evaluation report

---

### Objective

Maintain labelled cases and report retrieval quality, citation support, latency and cost.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 57: Cost estimate

---

### Objective

Separate fixed hosting costs from per-request model costs and document assumptions.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?



## Practical lab 58: Model card

---

### Objective

Describe intended use, data sources, evaluation, limitations and failure cases.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 59: Demo video

---

### Objective

Show a real query, retrieved evidence, cited answer, an unanswerable query and recovery.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 60: Resume evidence

---

### Objective

Map project features to real code, tests, metrics and your individual contribution.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

# Practical lab 61: PDF parsing

---

## Objective

Extract text and page metadata from public PDFs. Test empty, malformed and scanned documents; record limitations.

## Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

## Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

## Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

## Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 62: Chunk metadata

---

### Objective

Create stable chunk IDs with document, page, heading and index. Ensure every result traces to its source.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 63: Baseline retrieval

---

### Objective

Implement lexical retrieval on a fixed question set and report recall@k and missed cases.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 64: Vector retrieval

---

### Objective

Compare semantic search with lexical search using the same questions and relevance labels.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 65: Hybrid fusion

---

### Objective

Combine lexical and dense rankings; compare exact-term and semantic query types.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?



## Practical lab 66: Reranking

---

### Objective

Rerank candidate passages and measure relevance gains alongside latency and cost.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 67: Answer grounding

---

### Objective

Generate answers only from retrieved passages and validate citation IDs against known sources.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 68: Abstention

---

### Objective

Test questions that the corpus cannot answer; avoid fabricated facts or citations.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 69: Voice transcript

---

### Objective

Display raw speech transcript separately from normalized query and generated answer.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 70: Multilingual tests

---

### Objective

Test supported languages, code-switching, names, numbers and technical vocabulary.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

# Practical lab 71: Knowledge graph

---

## Objective

Define entity types, relationship types and source provenance; validate extracted edges.

## Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

## Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

## Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

## Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 72: Neo4j query

---

### Objective

Implement one multi-hop Cypher query and compare results against the no-graph baseline.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 73: API reliability

---

### Objective

Add validation, timeouts, structured errors, health checks and tests.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?



## Practical lab 74: Containerization

---

### Objective

Build a Docker image with external configuration and reproducible run instructions.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 75: Security review

---

### Objective

Review uploads, prompt injection, secrets, access control, logs and deletion policy.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 76: Evaluation report

---

### Objective

Maintain labelled cases and report retrieval quality, citation support, latency and cost.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 77: Cost estimate

---

### Objective

Separate fixed hosting costs from per-request model costs and document assumptions.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 78: Model card

---

### Objective

Describe intended use, data sources, evaluation, limitations and failure cases.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 79: Demo video

---

### Objective

Show a real query, retrieved evidence, cited answer, an unanswerable query and recovery.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 80: Resume evidence

---

### Objective

Map project features to real code, tests, metrics and your individual contribution.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

# Practical lab 81: PDF parsing

---

## Objective

Extract text and page metadata from public PDFs. Test empty, malformed and scanned documents; record limitations.

## Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

## Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

## Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

## Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?



## Practical lab 82: Chunk metadata

---

### Objective

Create stable chunk IDs with document, page, heading and index. Ensure every result traces to its source.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 83: Baseline retrieval

---

### Objective

Implement lexical retrieval on a fixed question set and report recall@k and missed cases.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 84: Vector retrieval

---

### Objective

Compare semantic search with lexical search using the same questions and relevance labels.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 85: Hybrid fusion

---

### Objective

Combine lexical and dense rankings; compare exact-term and semantic query types.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 86: Reranking

---

### Objective

Rerank candidate passages and measure relevance gains alongside latency and cost.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 87: Answer grounding

---

### Objective

Generate answers only from retrieved passages and validate citation IDs against known sources.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 88: Abstention

---

### Objective

Test questions that the corpus cannot answer; avoid fabricated facts or citations.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 89: Voice transcript

---

### Objective

Display raw speech transcript separately from normalized query and generated answer.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?



## Practical lab 90: Multilingual tests

---

### Objective

Test supported languages, code-switching, names, numbers and technical vocabulary.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

# Practical lab 91: Knowledge graph

---

## Objective

Define entity types, relationship types and source provenance; validate extracted edges.

## Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

## Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

## Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

## Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 92: Neo4j query

---

### Objective

Implement one multi-hop Cypher query and compare results against the no-graph baseline.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 93: API reliability

---

### Objective

Add validation, timeouts, structured errors, health checks and tests.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 94: Containerization

---

### Objective

Build a Docker image with external configuration and reproducible run instructions.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 95: Security review

---

### Objective

Review uploads, prompt injection, secrets, access control, logs and deletion policy.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 96: Evaluation report

---

### Objective

Maintain labelled cases and report retrieval quality, citation support, latency and cost.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 97: Cost estimate

---

### Objective

Separate fixed hosting costs from per-request model costs and document assumptions.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?



## Practical lab 98: Model card

---

### Objective

Describe intended use, data sources, evaluation, limitations and failure cases.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 99: Demo video

---

### Objective

Show a real query, retrieved evidence, cited answer, an unanswerable query and recovery.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

## Practical lab 100: Resume evidence

---

### Objective

Map project features to real code, tests, metrics and your individual contribution.

### Implementation plan

Define inputs and outputs; write down edge cases; implement the smallest useful version; add tests; run it in a clean environment; document what changed.

### Deliverables

A small code change, a test or benchmark, a README update, a short design note and a list of known limitations.

### Acceptance criteria

Another person can reproduce the result from the documented commands. Failures are handled safely and secrets are not exposed.

### Reflection

What failed? Which assumption was wrong? What would you improve? What evidence would justify adding complexity such as a graph database or another model?

# Engineering reference workbook 492: Responsible AI

---

## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 493: Data quality

---

## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 494: API design

---

## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 495: Observability

---

## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 496: Security

---

## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 497: Evaluation

---

## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 498: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 499: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 500: Knowledge graphs

---

## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 501: Deployment

---

## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 502: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 503: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 504: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 505: Data quality

---

## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 506: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 507: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 508: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 509: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 510: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 511: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 512: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 513: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 514: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 515: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 516: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 517: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 518: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 519: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 520: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 521: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 522: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 523: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 524: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 525: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 526: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 527: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 528: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 529: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 530: API design

---

## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 531: Observability

---

## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 532: Security

---

## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 533: Evaluation

---

## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 534: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 535: Voice UX

---

## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 536: Knowledge graphs

---

## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 537: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 538: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 539: Product thinking

---

## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 540: Responsible AI

---

## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 541: Data quality

---

## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 542: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 543: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 544: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 545: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 546: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 547: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 548: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 549: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 550: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 551: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 552: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 553: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 554: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 555: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 556: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 557: Evaluation

---

## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 558: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 559: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 560: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 561: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 562: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 563: Product thinking

---

## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 564: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 565: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 566: API design

---

## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 567: Observability

---

## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 568: Security

---

## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 569: Evaluation

---

## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 570: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 571: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 572: Knowledge graphs

---

## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 573: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 574: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 575: Product thinking

---

## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 576: Responsible AI

---

## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 577: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 578: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 579: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 580: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 581: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 582: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 583: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 584: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 585: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 586: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 587: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 588: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 589: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 590: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 591: Observability

---

## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 592: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 593: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 594: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 595: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 596: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 597: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 598: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 599: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 600: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 601: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 602: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 603: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 604: Security

---

## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 605: Evaluation

---

## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 606: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 607: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 608: Knowledge graphs

---

## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 609: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 610: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 611: Product thinking

---

## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 612: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 613: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 614: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 615: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 616: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 617: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 618: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 619: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 620: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 621: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 622: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 623: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 624: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 625: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 626: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 627: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 628: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 629: Evaluation

---

## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 630: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 631: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 632: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 633: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 634: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 635: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 636: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 637: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 638: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 639: Observability

---

## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 640: Security

---

## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 641: Evaluation

---

## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 642: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 643: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 644: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 645: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 646: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 647: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 648: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 649: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 650: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 651: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 652: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 653: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 654: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 655: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 656: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 657: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 658: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 659: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 660: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 661: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 662: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 663: Observability

---

## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 664: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 665: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 666: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 667: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 668: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 669: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 670: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 671: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 672: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 673: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 674: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 675: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 676: Security

---

## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 677: Evaluation

---

## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 678: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 679: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 680: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 681: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 682: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 683: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 684: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 685: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 686: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 687: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 688: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 689: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 690: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 691: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 692: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 693: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 694: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 695: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 696: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 697: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 698: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 699: Observability

---

## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 700: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 701: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 702: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 703: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 704: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 705: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 706: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 707: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 708: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 709: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 710: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 711: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 712: Security

---

## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 713: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 714: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 715: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 716: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 717: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 718: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 719: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 720: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 721: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 722: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 723: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 724: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 725: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 726: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 727: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 728: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 729: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 730: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 731: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 732: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 733: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 734: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 735: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 736: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 737: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 738: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 739: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 740: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 741: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 742: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 743: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 744: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 745: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 746: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 747: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 748: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 749: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 750: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 751: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 752: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 753: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 754: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 755: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 756: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 757: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 758: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 759: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 760: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 761: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 762: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 763: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 764: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 765: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 766: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 767: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 768: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 769: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 770: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 771: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 772: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 773: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 774: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 775: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 776: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 777: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 778: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 779: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 780: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 781: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 782: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 783: Observability

---

## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 784: Security

---

## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 785: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 786: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 787: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 788: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 789: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 790: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 791: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 792: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 793: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 794: API design

---

## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 795: Observability

---

## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 796: Security

---

## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 797: Evaluation

---

## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 798: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 799: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 800: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 801: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 802: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 803: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 804: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 805: Data quality

---

## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 806: API design

---

## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 807: Observability

---

## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 808: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 809: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 810: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 811: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 812: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 813: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 814: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 815: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 816: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 817: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 818: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 819: Observability

---

## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 820: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 821: Evaluation

---

## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 822: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 823: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 824: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 825: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 826: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 827: Product thinking

---

## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 828: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 829: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 830: API design

---

## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 831: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 832: Security

---

## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 833: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 834: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 835: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 836: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 837: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 838: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 839: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 840: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 841: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 842: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 843: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 844: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 845: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 846: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 847: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 848: Knowledge graphs

---

## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 849: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 850: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 851: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 852: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 853: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 854: API design

---

## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 855: Observability

---

## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 856: Security

---

## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 857: Evaluation

---

## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 858: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 859: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 860: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 861: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 862: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 863: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 864: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 865: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 866: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 867: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 868: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 869: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 870: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 871: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 872: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 873: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 874: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 875: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 876: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 877: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 878: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 879: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 880: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 881: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 882: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 883: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 884: Knowledge graphs

---

## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 885: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 886: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 887: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 888: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 889: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 890: API design

---

## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 891: Observability

---

## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 892: Security

---

## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 893: Evaluation

---

## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 894: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 895: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 896: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 897: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 898: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 899: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 900: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 901: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 902: API design

---

## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 903: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 904: Security

---

## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 905: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 906: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 907: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 908: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 909: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 910: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 911: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 912: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 913: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 914: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 915: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 916: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 917: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 918: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 919: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 920: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 921: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 922: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 923: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 924: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 925: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 926: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 927: Observability

---

## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 928: Security

---

## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 929: Evaluation

---

## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 930: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 931: Voice UX

---

## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 932: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 933: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 934: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 935: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 936: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 937: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 938: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 939: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 940: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 941: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 942: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 943: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 944: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 945: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 946: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 947: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 948: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 949: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 950: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 951: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 952: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 953: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 954: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 955: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 956: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 957: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 958: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 959: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 960: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 961: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 962: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 963: Observability

---

## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 964: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 965: Evaluation

---

## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 966: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 967: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 968: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 969: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 970: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 971: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 972: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 973: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 974: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 975: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 976: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 977: Evaluation

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## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 978: Retrieval

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## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 979: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 980: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 981: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 982: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 983: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 984: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 985: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 986: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 987: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 988: Security

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## Focus areas

secrets, authorization, prompt injection, upload limits, data isolation

## Practical assignment

Create or review a threat model for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 989: Evaluation

---

## Focus areas

labelled test sets, baselines, metrics, error analysis, reproducibility

## Practical assignment

Create or review a evaluation report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 990: Retrieval

---

## Focus areas

BM25, embeddings, fusion, metadata filters, reranking

## Practical assignment

Create or review a retrieval comparison for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you explain a project to a nontechnical person?

## Answer framework

Start with the user problem, demonstrate the workflow, report measured outcomes, explain risks and describe next steps plainly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 991: Voice UX

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## Focus areas

transcript correction, turn-taking, interruptions, latency, fallback

## Practical assignment

Create or review a voice test script for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is RAG?

## Answer framework

Retrieval-augmented generation retrieves external information and supplies it as context to a generator, helping ground answers without retraining the base model.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 992: Knowledge graphs

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## Focus areas

entity schemas, relationship validation, provenance, path queries

## Practical assignment

Create or review a graph schema for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

Why combine BM25 and vector search?

## Answer framework

Lexical retrieval helps exact terms, IDs and error codes; dense retrieval helps semantic matches. Hybrid search can improve coverage but must be measured.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.



# Engineering reference workbook 993: Deployment

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## Focus areas

Docker, environment config, health checks, rollback, resource limits

## Practical assignment

Create or review a deployment checklist for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you reduce unsupported answers?

## Answer framework

Improve retrieval, constrain generation, validate citations, evaluate faithfulness and abstain or clarify when evidence is insufficient.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 994: Career readiness

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## Focus areas

resume evidence, interview stories, job description mapping, demo quality

## Practical assignment

Create or review a application packet for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

What is reranking?

## Answer framework

A second-stage model reorders retrieved candidates for a query. It can improve relevance but adds latency and does not prove correctness.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 995: Product thinking

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## Focus areas

user needs, success metrics, constraints, usability, feedback loops

## Practical assignment

Create or review a product brief for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

When would you use Neo4j?

## Answer framework

When explicit entity relationships and multi-hop traversal solve a demonstrated problem that standard retrieval handles poorly.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 996: Responsible AI

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## Focus areas

privacy, harmful output handling, bias, transparency, limitations

## Practical assignment

Create or review a risk review for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you protect an API key?

## Answer framework

Keep it server-side in environment configuration or a secret manager; never commit it or expose it in browser code; rotate leaked keys.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 997: Data quality

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## Focus areas

schema validation, missing values, duplicates, freshness, lineage

## Practical assignment

Create or review a data quality report for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you evaluate a RAG system?

## Answer framework

Use a labelled test set and measure retrieval recall, citation support, answer correctness, abstention, latency and cost.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 998: API design

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## Focus areas

request schemas, status codes, retries, idempotency, error formats

## Practical assignment

Create or review a API contract for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How do you handle model provider failure?

## Answer framework

Use timeouts, safe retries for transient failures, rate limits, user-friendly errors and a tested fallback where appropriate.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Engineering reference workbook 999: Observability

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## Focus areas

structured logs, traces, latency percentiles, errors, token usage

## Practical assignment

Create or review a monitoring plan for VoiceRAG AI. Use a small reproducible example and write down assumptions before implementation.

## Technical review questions

What is the user-visible failure if this component is wrong? What test would detect it? Which data must remain private? What metric shows improvement? What fallback should users see?

## Interview drill

How would you test an AI agent?

## Answer framework

Validate tool schemas, restrict permissions, test malicious input, require confirmation for consequential actions and log tool calls.

## Deliverable checklist

■ Design note ■ Successful path ■ Failure cases ■ Reproducible test ■ README update ■ Limitations stated

## Career connection

Be ready to explain your own contribution, a trade-off, one measured result and one limitation. Do not present a tutorial exercise as production experience.

# Glossary, career checklists and official sources

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## Agent

A system that can select tools or steps to pursue a goal; it needs bounded permissions and evaluation.

## API

A defined interface for software components to exchange requests and responses.

## BM25

A lexical ranking method that scores documents based on query-term matching.

## Chunk

A document segment used as a retrieval unit.

## Embedding

A numeric vector representation of content used for similarity search.

## Hybrid search

A combination of lexical and semantic retrieval signals.

## LLM

Large language model trained to process and generate language.

## MLOps

Practices and infrastructure for deploying, monitoring and maintaining ML systems.

## RAG

Retrieval-augmented generation: retrieve relevant information and supply it as context to a generator.

## Reranker

A second-stage ranker that reorders retrieved candidates for a query.

## Recall@k

Fraction of relevant items found within the top k retrieved results.

## RRF

Reciprocal rank fusion, a method for combining ranked lists.

## Vector database

A system for storing and searching vector representations, often with metadata filters.

## WER

Word error rate, a common speech recognition metric.

## Career checklist

Verify each live role and location; tailor your resume honestly; link to code and a demo; prepare to explain design decisions; document limitations; never pay a recruiter for a job offer.

## Official careers source directory

See the company profile pages for individual official career URLs. Major sources include OpenAI, Anthropic, Google DeepMind, Google, Microsoft, NVIDIA, Meta, Amazon/AWS, Apple, IBM, Adobe, Salesforce, Databricks, Snowflake, Hugging Face, Neo4j, Pinecone, Qdrant, LangChain, LlamaIndex and others.

## Freshness note

Company products, role titles, hiring status, eligibility and interview stages change. This guide does not list verified live vacancies. Use official employer pages for current openings and role-specific requirements.

## Final reminder

Build a reliable text RAG system first, then add hybrid search, citations, voice, multilingual handling and graph retrieval only when evaluation justifies each layer.